NPTEL Environmental Science

IIT Kharagpur · Complete Assignments + Cheat Sheet + High-Chance Extras

200 assignment questions (12 weeks) **30** MSQs **117** high-likelihood predicted

Green = correct answer(s) **MSQ — select all** **best-guess** Week 2 keys **predicted** Practice only — not official

Cheat Sheet — Formulas, Must-Remember & Law Path

Formulas (24)

- **W1 · Arithmetic (linear) population growth:** $P_n = P_0 + nX$
- **W1 · Geometric population growth:** $P_n = P_0 (1 + r)^n$
- **W1 · Exponential population growth:** $P_n = P_0 e^(r n)$
- **W1 · IPAT environmental impact:** $I = P \times A \times T$
- **W1 · Ecological footprint of production:** $EF_{production} = P \times EQF / Y_w$
- **W1 · Energy Performance Index:** $EPI = E_{annual} / A_{floor}$
- **W5 · Betz limit / max wind power:** $C_{P,max}=16/27; P_{wind,max}= (8/27) \rho A v^3$
- **W5 · Power in the wind (kinetic flux):** $P_{wind} = (1/2) \rho A v^3$
- **W6 · Geothermal heat transfer:** $Q = \dot{m} * c_p * \Delta T$
- **W6 · Mean wave energy density:** $E = (1/8) \rho g H^2$
- **W6 · Deep-water wave power (per unit crest):** $P_W = (\rho g^2 H^2 T) / 32$
- **W6 · Hydropower potential energy / efficiency:** $PE = m g z$; $e_{hydro} = P_{elec} / (\dot{m} g z)$
- **W6 · Tidal energy resource idea:** Exploit tidal range (potential) or tidal stream KE when v >= 1.5-2 m/s
- **W10 · Sound pressure level:** $L_p = 20 \log10(P_{rms} / P_0)$, $P_0 = 20$ uPa
- **W10 · Sound wavelength:** $\lambda = c / f$
- **W10 · Inverse square law (point source):** $I = W/(4 \pi r^2)$; $L_{p2} = L_{p1} - 20 \log10(r2/r1)$
- **W10 · Combining equal sound levels:** Two equal incoherent sources -> L + 3 dB
- **W8 · NSF Water Quality Index (concept):** $WQI = \sum(Q_i * W_i)$ (normalized weights)
- **W11 · Bioconcentration Factor:** $BCF = C_{biota} / C_{water}$
- **W11 · Universal Soil Loss Equation:** $A = R K L S C P$
- **W6 · Fast pyrolysis (process note):** Biomass --(400-600 C, no O2)--> bio-oil + gas + char
- **W6 · EGS heat extraction concept:** Create permeability -> circulate fluid -> extract $Q=\dot{m} c \Delta T$ from hot rock
- **W11 · Logistic population growth:** $dN/dt = r N (1 - N/K)$
- **W11 · Octanol-water partition coefficient:** $K_{ow} = C_{octanol} / C_{water}$; log BCF ~ f(log K_{ow})

Week 12 Law Path

1. **Village gate -- Bishnoi and Khejri** — Community forest ethos; sacrifice legacy for trees precedes modern statutes.
2. **Himalayan trail -- Chipko 1973** — Tree-hugging movement; Bahuguna marches; Right Livelihood Award 1987.
3. **Western Ghats fork -- Appiko 1982-83** — Southern echo of Chipko forest protection.
4. **River bridge -- Water Act 1974** — CPCB/SPCBs; prevent/control water pollution; wholesomeness of water.
5. **Smokestack -- Air Act 1981 (am. 1987)** — Consent for listed industries; ambient pollutant limits.
6. **Umbrella pavilion -- EPA 1986 (am. 1991)** — Central powers to protect environment; parent of many Rules.
7. **Rule book stalls** — HW 1989, BMW 1998, Fly ash 1999, MSW 2000/2016, Batteries 2001, Plastic and E-waste 2011, C&D 2016.
8. **Green tribunal hall -- NGT 2010** — Natural justice; aim dispose in 6 months; Delhi + Bhopal, Pune, Kolkata, Chennai.
9. **Courtroom of PILs** — Landmark environmental judgments (incl. M.C. Mehta line) drive policy and education mandates.
10. **EIA desk at exit** — Rapid EIA case: predict impacts, consult, condition clearances before build.

Mnemonics

- **Energy trilemma SEE:** Security, Equity, Environmental sustainability *(Healthy grids must SEE clearly.)*
- **Waste hierarchy PRRRD:** Prevent > Reuse > Recycle > Recover > Dispose *(Please Reduce Really Revolting Dumps.)*
- **4Rs:** Reduce, Reuse, Recycle, Recover *(Source reduction sits at the throne.)*
- **IPAT:** Impact = Population x Affluence x Technology *(I Pat the planet harder as P, A, T rise.)*
- **Betz 16/27:** Max CP = 16/27 ~ 59% *(Sixteen over twenty-seven -- wind hard ceiling.)*
- **Five GHG names:** CO2, CH4, N2O, CFCs, H2O vapor *(Carbon Methane Nitro CFC Water -- trapping IR.)*
- **India Acts timeline W-A-E-N:** Water 1974, Air 1981, EPA 1986, NGT 2010 *(Wet Air envelops Environment; then green Tribunal.)*
- **USLE factors:** A = R K LS C P *(Rain Kills Loose Soil; Cover and Practice save it.)*
- **BCF trio:** Bioconcentration (vs water), Bioaccumulation (in body), Biomagnification (up food chain) *(Water -> Body -> Bigger predators.)*
- **Env-heavy SDGs:** 6 water, 7 energy, 11 cities, 12 SCP, 13 climate, 14 oceans, 15 land *(6-7-11-12-13-14-15 green spine.)*
- **Inverse square -6 dB:** Point source: double distance -> -6 dB *(Twice as far, six dB quieter.)*
- **NSF-WQI top weight:** DO weight 0.17 (highest) *(Dissolved Oxygen Dominates the index.)*
- **Ocean potential rank:** OTEC >> Wave >> Tidal (TWh/y) *(OTEC is an Ocean Titan; Tide is Tiny.)*
- **Linear vs Circular:** Take-Make-Waste vs Borrow-Use-Return *(Do not take forever -- borrow and bring back.)*

Compare hooks

Weather vs Climate: Weather is today's outfit; climate is the wardrobe over 30 years.

Geometric vs Exponential growth: Geometric steps yearly; exponential grows every instant -- e^(rn) wins.

Primary vs Secondary air pollutants: Primary exits the stack; secondary is cooked in sunlight.

Kyoto vs Paris: Kyoto bound Annex I; Paris asks every nation for NDCs.

Open vs Closed OTEC: Open boils seawater; closed boils ammonia-like fluid.

Point vs Line noise: Point loses 6 dB per double distance; road line loses ~3.

Act vs Rules: Act says WHAT; Rules say HOW.

Footprint vs Biocapacity: Footprint is shopping bill; biocapacity is paycheck from nature.

BOD vs COD: BOD = bugs oxygen appetite; COD = chemical oxidant appetite (usually >= BOD).

Mitigation vs Adaptation: Mitigation turns down the stove; adaptation learns to live with the heat.

Week 1 — Environmental Concerns, Population Growth and Resource Consumption

- IPAT: I=P x A x T -- know what A and T can represent
- Geometric != exponential: e^(rn) grows faster than (1+r)^n for same r
- EF units = global hectares; include carbon land
- EPI units = kWh/m^2-year
- Three pillars: environment, economy, society
- Of 30 most polluted cities, ~21 in India (course claim)
- *Pn = P0 + nX (arithmetic/linear)*
- *Pn = P0(1+r)^n geometric vs Pn = P0 e^(rn) exponential*
- *IPAT: I = P x A x T (Ehrlich and Holdren, 1971)*
- *Ecological footprint in global hectares (gha); biocapacity vs demand*
- *Earth Overshoot Day: demand exceeds year's biocapacity*
- **World population (UNPF, 15 Nov 2022):** 8.0 billion — Baseline for IPAT/population questions
- **India population (Census 2011):** 1.21 billion — Course IPAT numerical examples
- **India avg annual pop growth (census fit):** ~1.75%/y total; ~2.85%/y urban — Curve-fitting / exponential model
- **Global EF vs biocapacity per person:** 2.8 vs 1.6 gha — Overshoot Day rationale

Week 2 — Principles of Sustainable Development

- Memorize Brundtland one-liner verbatim
- Timeline: 1987 Brundtland -> 1992 Rio -> 2000 MDGs -> 2015 SDGs
- Nine planetary boundaries concept (names > exact numbers)
- SDG #6 water, #7 energy, #11 cities, #12 SCP, #13 climate -- frequent env links
- Sustainability requires all three pillars, not environment alone
- *Brundtland 1987: meets present needs without compromising future generations*
- *Rio Earth Summit 1992 adopted sustainable development agenda*
- *MDGs 2000 -> SDGs 2015 (17 goals, 2030 Agenda)*
- *Nine planetary boundaries; safe operating space (Rockstrom/Steffen)*
- *Anthropocene Great Acceleration in socio-economic and Earth-system indicators*
- **Brundtland report:** 1987 — Definition origin year
- **Rio Earth Summit:** 1992 — SD global adoption
- **SDGs adopted:** 2015; 17 goals — High-frequency MCQ
- **Planetary boundaries papers:** Rockstrom 2009; Steffen 2015 — Cite authors/years

Week 3 — Understanding Air Pollution

- Definition = concentration OR duration causing harm/interference
- Primary/secondary and stationary/mobile/area -- two orthogonal classifications
- Tropospheric O3 is secondary; stratospheric O3 depletion is different
- Montreal 1987 != climate Kyoto/Paris
- Coal S -> SO2 emission stoichiometry is a recurring numerical
- *Ambient air = outdoor atmospheric air supporting life*
- *Primary vs secondary pollutants (e.g., O3, secondary PM)*
- *Criteria air pollutants vs hazardous/non-criteria*
- *Sources: stationary, mobile, area*
- *ESP / baghouse / scrubbers / FGD control options*
- **Industrial particulate control efficiency:** Often 95-99% — ESP/baghouse MCQs
- **FGD limestone stoichiometry example:** ~1.03 (3% excess) — Scrubber design calculation
- **Indoor air pollution deaths (cited):** ~3.2 million/year (2020) — Health burden fact
- **Montreal Protocol:** 1987 — Ozone policy date

Week 4 — Climate Change: Scientific Basis, Mitigation and Adaptation

- Do not confuse weather with climate
- Five major GHG contributors: CO2, CH4, N2O, CFCs, H2O
- GWP(CO2)=1 always
- Timeline: UNFCCC 1992 -> Kyoto 1997 -> Paris 2015
- IPCC provides assessments; Nobel 2007 with Al Gore
- *Weather = short-term; climate = long-term statistics (decades+)*
- *GHGs: CO2, CH4, N2O, CFCs, H2O vapor -- transparent to SW, trap IR*
- *GWP: CO2 = 1 by definition*
- *Enhanced greenhouse effect from anthropogenic GHG buildup*
- *Albedo feedback: ice loss -> lower albedo -> more warming*
- **Warming rate since 1981 (course):** ~0.18 C per decade — Evidence of acceleration
- **Kyoto Annex I target:** 5.2% below 1990 — Treaty number
- **Doha Amendment:** >=18% below 1990 by 2020 — Second commitment period
- **Sept 2012 Arctic sea ice:** Record low; ~44% below long-term — Indicator fact

Week 5 — Renewable Energy I (Solar and Wind Energy)

- SEE trilemma mnemonic: Security, Equity, Environment
- P proportional to v^3 -- doubling wind speed -> 8x power
- Pwind,max = (8/27) rho A v^3
- CP is <= Betz; real turbines lower due to losses + CF
- Renewables defined by replenishment rate, not zero impact
- *Renewable = replenished at rate >= rate of use*
- *Energy trilemma: Security, Equity, Environmental sustainability (SEE)*
- *Incoming solar ~5.4 million EJ/year*
- *Renewables ~12.6% of total final energy consumption (2020, course)*
- *Betz limit CP,max = 16/27 ~ 59.3%*
- **Betz limit:** 16/27 ~ 59.3% — Must-memorize wind fact
- **Air density (example):** 1.21 kg/m^3 — Wind power numericals
- **Global solar radiation input:** ~5.4e6 EJ/y — Resource scale
- **RE share of TFEC 2020:** ~12.6% (from 8.7% in 2009) — Status MCQ

Week 6 — Renewable Energy II (Geothermal, Ocean, Hydro, Bioenergy)

- Rank ocean potentials: OTEC >> wave >> tidal (as taught)
- Hydro efficiency uses head z, not kinetic of river alone
- EGS = engineered permeability in hot dry rock
- Wave power deep water proportional to H^2 T
- Pyrolysis != combustion: no/limited O2
- *Geothermal origins: primordial heat + radiogenic decay + solar-heated shallow ground*
- *EGS: stimulate hot dry rock / low-permeability reservoirs*
- *Geothermal heat Q = m c DeltaT (district heating example)*
- *Tidal potential ~1200 TWh/y (smallest ocean); OTEC ~44000; wave ~29500 TWh/y*
- *Tidal stream useful when flow ~>= 1.5-2 m/s; spring vs neap tides*
- **Geothermal example heat:** 42 TJ/y (75 C, 20 L/s, 170 d, DeltaT=35 C) — Q=mcDeltaT worked example
- **Tidal stream speed threshold:** 1.5-2 m/s — Resource screening
- **Hydro example:** 150 MW, z=35 m, Q=500 m^3/s -> compute ehydro — Efficiency numerical pattern
- **Fast pyrolysis temperature:** 400-600 C — Bioenergy process ID

Week 7 — Industrial Ecology and Circular Economy

- Anabolism/catabolism metaphor for cities
- Linear vs circular slogans exactly
- Industrial symbiosis = waste -> feedstock
- Circular economy supports SDG12 (responsible consumption/production)
- *Urban metabolism = anabolism + catabolism of cities*
- *Linear = Take-Make-Waste; Circular = Borrow-Use-Return*
- *Closing the loop reduces virgin extraction and waste emissions*
- *Industrial ecology: industry as ecosystem with symbiotic flows*
- *Linear throughput runs down the parent ecosystem*
- **Conceptual shift emphasized:** From linear urban subsystem to closed-loop economy — Essay/short-note framing
- **Core CE strategies:** Reuse, remanufacture, recycle, design out waste — List for 4-5 markers

Week 8 — Water Quality, Reuse and Recycling

- Withdrawals != consumption
- BOD3 @ 27 C is Indian convention (not always BOD5 @ 20 C)
- Highest NSF weight = DO (0.17)
- NGT principal bench New Delhi; 4 other places
- Point vs non-point pollution sources
- *Salt water ~97.5%; freshwater ~2.5%; usable freshwater tiny fraction*
- *Withdrawals = consumption + returns; municipal WW often ~80% of water use (India cities ~87%)*
- *DO, BOD (3 d @ 27 C India), COD, N species key parameters*
- *DO sag / oxygen deficit from BOD loading*
- *NSF-WQI: weighted Qi; DO weight 0.17, fecal coliform 0.16*
- **World salt vs fresh water:** 97.5% vs 2.5% — Scarcity framing
- **Municipal WW as % of water use:** ~80% typical; ~87% major Indian cities — Mass-balance intuition
- **India household use (cited):** ~100 L/cap-d — Design/demand MCQ
- **NGT established:** 18 Oct 2010 — Institution date

Week 9 — Solid Waste Management (including E-waste)

- Hierarchy order beats technology name-dropping
- Methane + leachate = twin landfill hazards
- VS/TS ratio guides biological treatment
- E-waste needs EPR and authorized channels
- Source reduction > end-of-pipe
- *Waste hierarchy: Prevent > Reuse > Recycle > Recover > Dispose*
- *4Rs: Reduce, Reuse, Recycle, Recover (source reduction first)*
- *Solid waste includes containerized liquid/gas wastes in definition*
- *Open dumps -> air (VOCs, CH4 explosions), water, soil contamination*
- *Moisture, TS, VS, FS characterization maths*
- **Moisture example:** 40% moisture -> TS=60%; if 50/60 lost on ignition -> VS=83.3% of TS — Characterization numerical
- **MSW Rules evolution:** 2000 -> revised 2016 — Legislation link to W12
- **Open dump hazards:** Air, SW/GW, soil, methane explosion risk — Why hierarchy matters

Week 10 — Noise Pollution

- lambda=c/f and Lp=20 log(P/P0)
- -6 dB rule for point sources when distance doubles
- +3 dB for two identical sources
- A-weighting for most environmental limits
- Noise is energy pollution -- hearing damage can be permanent
- *lambda = c/f ; period p=1/f*
- *Lp = 20 log(Prms/P0); P0=20 uPa = 0 dB*
- *c ~ 20.05*sqrt(T) (T in K)*
- *I = W/(4 pi r^2) inverse square; Delta Lp ~ -20 log(r2/r1) -> -6 dB per distance doubling*
- *Line source: intensity proportional to 1/r (not 1/r^2)*
- **Reference pressure:** 20 uPa — 0 dB definition
- **Two equal sources:** +3 dB — Combination rule
- **Occupational caution:** >=85 dB(A) over 8 h risky — Health standard
- **Conversation / pain scale anchors:** ~60 dB converse; ~140 dB pain/jet — Order-of-magnitude

Week 11 — Ecology and Soil Pollution

- BCF = Cbiota/Cwater
- USLE letters: R K L S C P
- Energy one-way; nutrients cycle
- Biomagnification != bioaccumulation (know the difference)
- Carrying capacity converts exponential -> logistic
- *Ecology from oikos = house; study of organisms at home*
- *Energy flow unidirectional; mass/nutrients cyclic*
- *Trophic levels: producers -> herbivores -> carnivores*
- *Logistic growth: exponential modified by carrying capacity K*
- *BCF = Cbiota/Cwater (L/kg); correlates with Kow*
- **DDT BCF example (gull):** up to ~120,000x vs water — Biomagnification shock value
- **Solar spectrum note:** ~99% energy 136-4000 nm; ~7% UV, ~46% IR — Energy ecology
- **USLE unit plot:** 5.5 ft x 72.6 ft, 9% slope — Definition of factors
- **Incident solar at autotrophs:** ~20% of total incident (course) — Energy pyramid setup

Week 12 — Environmental Policies and Legislation

- Sequence: Water 1974 -> Air 1981 -> EPA 1986 -> NGT 2010
- Act vs Rules distinction
- Chipko 1973; Appiko 1982-83; Bishnoi Khejri legacy
- MSW Rules 2000 revised 2016; E-waste/Plastic Rules 2011
- NGT not bound by CPC 1908; natural justice; try dispose in 6 months
- *Movements: Bishnoi (Khejri), Chipko (1973), Appiko (1982-83), Jungle Bachao*
- *Water Act 1974; Air Act 1981 (am. 1987); EPA 1986 (am. 1991)*
- *Act = law/substance; Rules = procedure under the Act*
- *HW Rules 1989; BMW 1998; MSW 2000/2016; Batteries 2001; Plastic/E-waste 2011; C&D 2016*
- *NGT Act 2010; principles of natural justice; 6-month disposal endeavor*
- **Water Act:** 1974 — First major pollution Act
- **Air Act:** 1981 (amended 1987) — Date trap
- **EPA:** 1986 (amended 1991) — Umbrella year
- **NGT:** 2010 Act; est. 18.10.2010 — Institution

Assignments — All 12 Weeks (200 questions)

Week 1 — Environmental Concerns, Population & Resources

20 questions · 6 MSQ · 14 other

Q1 MCQ — Environmental Concerns, Population & Resources

A chemical is found in a river at concentrations above natural background levels. No adverse ecological effects are observed. The chemical is best classified as:

A. Pollutant only

B. Contaminant only

C. Both pollutant and contaminant

D. Neither

Answer: B

Why: Contamination means the compound is present at concentrations above background level; Pollution means the compound causes adverse effect on human or ecological systems.

Q2 MCQ — Environmental Concerns, Population & Resources

According to the Brundtland Report, sustainable development requires:

A. Meeting present needs without compromising future generations needs

B. Unlimited resource extraction for economic growth

C. Balancing environmental protection with quality-of-life improvement

D. Long-term protection of environmental systems

Answer: A

Why: Development should satisfy current needs while preserving the ability of future generations to satisfy theirs, that means-Present generation must benefit (quality of life); Future generations must not suffer; and Environment must be protected Option B violates sustainability because unlimited extraction leads to resource depletion.

Q3 MCQ — Environmental Concerns, Population & Resources

A researcher obtains the following R^2 values while fitting population data: Model A: 0.32 Model B: 0.71 Model C: 0.96 Which model best describes the observed data?

A. Model A

B. Model B

C. Model C

D. Cannot be determined

Answer: C

Why: R^2 measures goodness of fit. R2 >0.9, it is a reasonably good fit and >0.95 is a very good fit. Values of R2 <0.5 show that there is no correlation between the two independent variables.

Q4 MCQ — Environmental Concerns, Population & Resources

If a class has a capacity for 100 students now, and the capacity for students is to be increased by 10% every year, what will the student intake capacity be at the end of 30 years?

A. 2718

B. 2008

C. 1500

D. 3000

Answer: B

Why: As P0 = 100, n = number of years = 30 and r = 10% or 0.1 P=P0*exp (n*r) = 100*exp (30*0.1) = 2008

Q5 MCQ — Environmental Concerns, Population & Resources

A country successfully reduces its population growth rate but doubles its per-capita resource consumption. What is the most likely outcome?

A. Environmental pressure will certainly decrease.

B. Environmental pressure may still increase.

C. Resource consumption becomes irrelevant.

D. Pollution must decrease.

Answer: B

Why: Environmental impact depends on: Population Consumption Technology So, if population growth decreases, a large increase in consumption can offset the gains. This is why some low-population countries still have very high environmental footprints.

Q6 MSQ — select all — Environmental Concerns, Population & Resources

Which of the following are correctly classified as renewable resources?

A. Air

B. Soil

C. Coal

D. Water

Answer: A, B, D

Why: renewable resource can naturally restore within a reasonable time scale. Air is continuously renewed through atmospheric processes. Water is renewed through the hydrologic cycle.

Q7 MCQ — Environmental Concerns, Population & Resources

Organic matter discharged into a river is decomposed/degraded naturally. This ability or self-purification capacity of a river is an example of:

A. Natural resource

B. Natural service

C. Non-renewable resource

D. Green design

Answer: B

Why: The river is performing an ecological function. The water itself is a resource. Its ability to assimilate and degrade wastes is a service provided by nature.

Q8 MCQ — Environmental Concerns, Population & Resources

A city imports food, fuel, and manufactured products while exporting large amounts of waste. This city most closely resembles:

A. Agropolis

B. Petropolis

C. Ecopolis

D. Circular city

Answer: B

Why: Petropolis is characterized by: Dependence on fossil fuels, Heavy transportation requirements, Import of goods and services Export of wastes

Q9 MCQ — Environmental Concerns, Population & Resources

Two buildings use identical construction materials. Building X minimizes energy use only during operation. Building Y minimizes energy and resource use throughout manufacturing, operation, maintenance, and disposal. Which is more consistent with sustainable design?

A. Building X

B. Building Y

C. Both equally

D. Neither

Answer: B

Why: Green design mostly focuses on reducing current environmental impacts. Sustainable design considers the entire lifecycle: Manufacturing, Transportation, Operation, Maintenance, Disposal. Therefore, Building Y represents sustainable design.

Q10 MCQ — Environmental Concerns, Population & Resources

What is the fundamental weakness of a linear economy?

A. Resources are endlessly recycled.

B. Waste generation is minimized.

C. Resources move in a one-way flow from extraction to disposal.

D. Renewable energy dominates production.

Answer: C

Why: Linear Economy: Take Make Dispose Resources are extracted, converted into products, and discarded. This model ignores material recovery.

Q11 MSQ — select all — Environmental Concerns, Population & Resources

Which of the following changes would move a city from a Petropolis toward an Ecopolis?

A. Increasing local food production

B. Expanding waste recycling

C. Increasing fossil-fuel imports

D. Promoting renewable energy

Answer: A, B, D

Why: An Ecopolis: Reduce external dependence, minimize waste, Close material loops, use renewable resources so, Local production, Recycling and Renewable energy support the transition. Increasing fossil-fuel imports does the opposite.

Q12 MCQ — Environmental Concerns, Population & Resources

If a country doubles its electricity generation using coal while maintaining the same population. What is the most likely environmental consequence?

A. Reduced greenhouse gas emissions

B. Increased resource consumption and pollution

C. No environmental change

D. Reduced dependence on natural resources

Answer: B

Why: Coal: Non-renewable and Carbon intensive Doubling coal-based generation implies: More coal extraction More combustion More CO emissions More air pollution Greater environmental burden. Thus, environmental impacts increase even without population growth.

Q13 MCQ — Environmental Concerns, Population & Resources

The Capital Stock Model suggests that sustainable development is achieved when:

A. Capital stock is continuously depleted

B. Natural resources are exhausted

C. Society lives on interest rather than capital

D. Population is reduced to zero

Answer: C

Why: The lecture compares sustainability to living on the interest generated by capital rather than consuming the capital itself. Consuming the capital stock threatens long- term prosperity.

Q14 MCQ — Environmental Concerns, Population & Resources

Population = 200 million, Per capita electricity use = 1500 kWh/person-year, CO2 Emission factor = 0.9 kg/kWh, Calculate Total annual CO emissions?

A. 180*10^9 kg

B. 270*10^9 kg

C. 300*10^9 kg

D. 450*10^9 kg

Answer: B

Why: 200*(10^6) * 1500*0.9=270 (10^9) kg

Q15 MSQ — select all — Environmental Concerns, Population & Resources

Which of the following are pillars of sustainable development?

A. Social

B. Ecological

C. Financial

D. Military

Answer: A, B, C

Why: The three pillars correspond to social, environmental, and economic/financial dimensions.

Q16 MSQ — select all — Environmental Concerns, Population & Resources

Which represent ecological deficit?

A. EF=3, BC=2

B. EF=2, BC=3

C. EF=5, BC=4

D. EF=4, BC=4

Answer: A, C

Why: Deficit occurs when: EF>BC

Q17 MCQ — Environmental Concerns, Population & Resources

Earth Overshoot Day represents:

A. The day Earth begins regeneration

B. The day humanity's resource demand exceeds Earth's annual regeneration capacity

C. The first day of the year

D. The day carbon emissions become zero

Answer: B

Why: Earth Overshoot Day is the date when humanity has consumed all ecological resources that Earth can regenerate during that year. After this day, humanity is operating on an ecological deficit.

Q18 MCQ — Environmental Concerns, Population & Resources

A building consumes 60,000 kWh/year, Area = 1,500 m, Calculate EPI.

A. 20 kWh/m2-year

B. 30 kWh/m2-year

C. 40 kWh/m2-year

D. 50 kWh/m2-year

Answer: C

Why: EPI = [Annual Energy Consumption/Floor Area] = [60000/1500] = 40 kWh/m2-year

Q19 MSQ — select all — Environmental Concerns, Population & Resources

Factors affecting EPI include:

A. Number of occupants

B. Climate and Seasonal Variation

C. Energy conservation practices

D. Shape and building orientation

Answer: A, B, C, D

Why: Number of Occupants: More occupants generally increase lighting demand, cooling load, appliance usage. Climate and Seasonal Variations: Hot climates increase air-conditioning demand, energy consumption. Cold climates increase heating demand.

Q20 MSQ — select all

Environmental Concerns, Population & Resources

Which paper products generate more than 8,000 kg GHG emissions?

A. Toilet Paper

B. Uncoated A4 Paper

C. Paperboard (uncoated, unbleached, Kraft)

D. Newsprint (uncoated ground wood)

Answer: A, B, D

Why: Toilet paper = 11,897 A4 = 8,999 Paperboard = 1,495 [It is made from 100% recycled fibre, using recycled fibres eliminates many energy-intensive steps associated with producing virgin pulp. It does not require virgin wood, so no fresh trees are harvested, emissions from logging, transportation, and initial wood processing are avoided.] Newsprint = 9,681

Week 2 — Sustainable Development

10 questions · 0 MSQ · 10 other · best-guess keys

Q1 MCQ best-guess — Sustainable Development

According to the Planetary Boundaries Framework, several Earth-system processes have already exceeded their safe operating limits. Which of the following boundaries is considered to have been crossed?

A. Ocean acidification

B. Stratospheric ozone depletion

C. Fresh water use

D. Land system change

Answer: D

Why: No official answer key in source. Land-system change is among boundaries assessed as crossed; ocean acidification, ozone, and freshwater were generally within safer ranges at the time of the framework updates taught in the course.

Q2 MCQ best-guess — Sustainable Development

The Brundtland Commission was established primarily to address growing concerns about

A. Population growth exceeding carrying capacity

B. Widening disparities between economic growth and environmental protection

C. Global financial instability and trade imbalances

D. Depletion of fossil fuel reserves

Answer: B

Why: No official answer key. The Brundtland Commission was convened to address the growing gap between economic growth and environmental protection.

Q3 MCQ best-guess — Sustainable Development

Which of the following is the best example of a negative international spillover effect in the context of sustainable development?

A. Relocation of pollution-intensive industries from developed to developing countries

B. Transfer of clean-energy technologies to developing nations

C. Adoption of renewable energy technologies leading to reduced transboundary air pollution

D. International cooperation for the conservation of biodiversity

Answer: A

Why: No official answer key. Relocating pollution-intensive industries transfers environmental burdens across borders (negative spillover).

Q4 MCQ best-guess — Sustainable Development

Which of the following Sustainable Development Goals (SDGs) is most directly focused on promoting peace, justice, strong institutions, accountability, and transparent governance?

A. SDG #8

B. SDG #10

C. SDG #16

D. SDG #13

Answer: C

Why: No official answer key. SDG 16 focuses on peace, justice, and strong institutions.

Q5 MCQ best-guess — Sustainable Development

The inclusion of "Partnership" as one of the five "P's" of the SDGs underscores the importance of

A. international cooperation and multi-stakeholder collaboration for sustainable development

B. mobilizing financial resources exclusively for developed countries

C. promoting investment primarily in developed economies

D. privatizing environmental governance and public services

Answer: A

Why: No official answer key. Partnership among the 5 Ps emphasizes multi-stakeholder and international collaboration.

Q6 MCQ best-guess — Sustainable Development

What is the primary significance of the SDG Index as a tool for assessing sustainable development performance?

A. It focuses exclusively on environmental indicators

B. It ranks countries based on geopolitical influence

C. It replaces national statistical systems for development monitoring

D. It offers a composite measure of progress toward achieving the SDGs.

Answer: D

Why: No official answer key. The SDG Index is a composite score of country progress toward the SDGs.

Q7 MCQ best-guess — Sustainable Development

Which of the following best reflects the primary objective of Smart City initiatives?

A. To achieve full automation of urban services through artificial intelligence.

B. To minimize government involvement by promoting privately managed urban systems.

C. To enhance urban sustainability, efficiency, and quality of life through the integration of digital technologies and data-driven governance

D. To replace conventional infrastructure with futuristic urban designs and architecture.

Answer: C

Why: No official answer key. Smart City initiatives aim to improve sustainability, efficiency, and quality of life via digital/data-driven governance.

Q8 MCQ best-guess Sustainable Development

Which of the following statements about extensive green roofs is correct?

A. They use deep layers of soil to grow large plants.

B. They require frequent watering.

C. They are significantly heavier than intensive green roofs.

D. They use a thin layer of soil and feature easy-to-maintain vegetation.

Answer: D

Why: No official answer key. Extensive green roofs use a thin soil layer and low-maintenance vegetation.

Q9 MCQ best-guess Sustainable Development

Which of the following is a significant environmental impact of the urban heat island effect?

A. Reduced temperature differences between urban and rural areas

B. Increased energy consumption for cooling in cities.

C. Enhanced groundwater recharge through increased infiltration.

D. Reduced generation of anthropogenic heat from human activities.

Answer: B

Why: No official answer key. Urban heat islands raise cooling energy demand in cities.

Q10 MCQ best-guess Sustainable Development

Which of the following characteristics most clearly distinguishes sustainable urbanization from conventional patterns of urban growth?

A. Dependence on private automobiles for mobility

B. Emphasis on low-density suburban development

C. Integration of environmental resilience, social inclusivity, and resource efficiency

D. Strict separation of residential and commercial land uses

Answer: C

Why: No official answer key. Sustainable urbanization integrates environmental resilience, social inclusivity, and resource efficiency.

Week 3 — Air Pollution

15 questions · 5 MSQ · 10 other

Q1 MCQ — Air Pollution

Which of the following is an example of a secondary air pollutant, formed in the atmosphere through chemical reactions involving primary pollutants?

A. Volatile organic compounds

B. Sulphur dioxide

C. Ground-level ozone

D. Carbon monoxide

Answer: C

Why: Ground-level ozone (O3) is a secondary air pollutant because it is formed in the atmosphere through reactions between nitrogen oxides (NO) and volatile organic compounds (VOCs) in the presence of sunlight. The other options are primary pollutants emitted directly from sources.

Q2 MCQ — Air Pollution

Which of the following atmospheric conditions is most conducive to the formation of a radiation inversion?

A. Cloudy conditions that limit nighttime heat loss

B. Intense daytime heating of the Earth's surface

C. Rainfall associated with monsoon weather systems

D. Clear, calm nights that promote rapid surface cooling.

Answer: D

Why: Radiation inversions commonly occur on clear, calm nights when the ground loses heat rapidly through radiation. This cools the air near the surface more than the air above, creating a temperature inversion in which temperature increases with height.

Q3 MCQ — Air Pollution

Which fraction of particulate matter can penetrate deepest into the human body?

A. PM10

B. PM2.5

C. PM0.1

D. Total Suspended Particle

Answer: C

Why: Ultrafine particles (PM0.1) can translocate to multiple organs and cause severe health impacts.

Q4 MCQ — Air Pollution

Which characteristic of fine particulate matter (PM2.5) enables it to be the dominant component of transboundary haze?

A. They are largely insoluble in water.

B. They can remain suspended in the atmosphere for extended periods and be transported over long distances.

C. They have a short atmospheric residence time.

D. They are transported only under conditions of strong winds.

Answer: B

Why: Fine particles (PM2.5) have very low settling velocities, allowing them to remain airborne for days to weeks. As a result, they can be carried over hundreds or even thousands of kilometers by atmospheric circulation, making them a major contributor to transboundary haze.

Q5 MCQ — Air Pollution

In a dry atmosphere, under which condition is the atmosphere considered unstable?

A. The dry adiabatic lapse rate is greater than the environmental lapse rate.

B. The dry adiabatic lapse rate is equal to the environmental lapse rate.

C. The dry adiabatic lapse rate is less than the environmental lapse rate.

D. The moist adiabatic lapse rate is equal to the environmental lapse rate.

Answer: C

Why: An atmosphere is unstable when the environmental lapse rate (ELR) exceeds the dry adiabatic lapse rate (DALR). In this case, a rising air parcel remains warmer and less dense than its surroundings, causing it to continue rising. Therefore, atmospheric instability occurs when the DALR is less than the ELR.

Q6 MCQ — Air Pollution

Selective Catalytic Reduction (SCR) technology is an air pollution control technology primarily used to reduce emissions of _____, converting them into harmless _____ and _____.

A. SOx, S2, and H2

B. NOx , N2, and H2O

C. CO2 , C, and H2

D. VOCs, N2, and H2O

Answer: B

Why: Selective catalytic reduction (SCR) is an advanced technology used to control nitrogen oxide (NO) emissions from diesel engines. In this process, a diesel exhaust fluid (DEF), typically a urea-based solution, is injected into the exhaust stream upstream of a catalyst. The heat of the exhaust converts the DEF into ammonia (NH3), which then reacts with nitric oxide (NO) and nitrogen dioxide (NO2) on the surface of the catalyst.

Q7 MCQ — Air Pollution

A region is exposed to severe acid deposition over an extended period. Which of the following environmental impacts is most likely to occur first?

A. Complete disappearance of forest vegetation.

B. Increase in atmospheric ozone concentration.

C. Depletion of soil buffering capacity due to the loss of Ca2 and Mg2 ion.

D. Enhanced carbon sequestration in terrestrial ecosystems.

Answer: C

Why: Acid deposition leaches essential base cations such as calcium (Ca2) and magnesium (Mg2) from soils, reducing their ability to neutralize acidity. This loss of buffering capacity is typically one of the earliest environmental effects and can eventually lead to soil degradation, forest decline, and aquatic ecosystem damage.

Q8 MCQ — Air Pollution

Which of the following compounds is not considered an ozone-depleting substance (ODS)?

A. Methyl fluoride

B. Methyl chloride

C. Methyl bromide

D. Methyl chloroform

Answer: A

Why: Ozone depletion is primarily caused by compounds containing chlorine or bromine, which release reactive halogen radicals in the stratosphere. Methyl fluoride (CH3F) contains fluorine but no chlorine or bromine and has negligible ozone-depleting potential. In contrast, methyl chloride, methyl bromide, and methyl chloroform are recognized ozone-depleting substances.

Q9 MCQ — Air Pollution

A dry air parcel with an initial temperature of 30�C rises 1 km in the atmosphere. Assuming a dry adiabatic lapse rate (DALR) of 9.8�C km�1 and an environmental temperature of 22�C at 1 km altitude, what will most likely happen to the parcel?

A. It will continue rising.

B. It will become neutrally buoyant.

C. It will sink back toward the surface.

D. It will form a temperature inversion.

Answer: C

Why: The parcel cools at the DALR as it rises: Tparcel = 30-9.8 = 20.2�C At 1 km, the surrounding air temperature is 22�C, which is warmer than the parcel (20.2�C). Since the parcel is cooler and therefore denser than its surroundings, it experiences negative buoyancy and tends to sink back toward the surface. Thus, the atmosphere is stable under these conditions.

Q10 MCQ — Air Pollution

The sub-index values of NO, SO2, and PM10 are 60, 90, and 40, respectively. According to the National Air Quality Index (NAQI) released by the Government of India in the year 2015, the overall NAQI is

- **A.** 80
- **B.** 200
- **C.** 67
- **D.** 70

Answer: A

Why: According to NAQI, overall AQI equals the highest pollutant sub-index. With NO=60, SO2=90, PM10=40, NAQI=90. (Source answer key lists 'a. 90'; option labels in the text appear misaligned.)

Q11 MSQ — select all — Air Pollution

Which of the following factors directly influence the atmospheric dispersion of pollutants?

- **A.** Wind speed
- **B.** Atmospheric stability
- **C.** Relative humidity
- **D.** Emission height

Answer: A, B, D

Why: Atmospheric dispersion is affected by several meteorological and source-related factors. Wind speed controls the rate of pollutant transport and dilution, atmospheric stability determines the extent of vertical mixing, and emission height influences the distance pollutants travel before reaching ground level.

Q12 MSQ — select all — Air Pollution

Which of the following statements about ozone-depleting substances (ODSs) are correct?

- **A.** Chlorofluorocarbons (CFCs) are ODSs.
- **B.** ODSs are highly reactive in the troposphere.
- **C.** Halons are ODSs.
- **D.** ODSs contribute to the destruction of stratospheric ozone.

Answer: A, C, D

Why: CFCs and halons are well-known ODSs. After reaching the stratosphere, they release chlorine and bromine radicals that catalytically destroy ozone. Most ODSs are actually chemically stable in the troposphere, which allows them to persist long enough to reach the stratosphere.

Q13 MSQ — select all — Air Pollution

Which of the following are the primary precursors involved in the formation of photochemical smog?

A. aldehydes

B. nitrogen oxides

C. peroxyacetyl nitrate

D. volatile organic compounds

Answer: B, D

Why: Photochemical smog forms when nitrogen oxides (NO) and volatile organic compounds react in the presence of sunlight. Aldehydes and PAN are secondary pollutants produced during these photochemical reactions.

Q14 MSQ — select all — Air Pollution

Which of the following health effects may result from prolonged exposure to indoor air pollution?

A. Asthma exacerbation

B. Chronic obstructive pulmonary disease (COPD)

C. Lung cancer

D. Cardiovascular diseases

Answer: A, B, C, D

Why: From assignment solution.

Q15 MSQ — select all — Air Pollution

A pollutant is emitted at the same rate from two stacks: • Stack A: 80 m height • Stack B: 250 m height An atmospheric temperature inversion extends from the ground surface up to 200 m. Which of the following statements are most likely correct?

A. Pollutants from Stack A are more likely to be trapped below the inversion.

B. Pollutants from Stack B may disperse more effectively.

C. Pollutants from Stack B are more likely to be trapped below the inversion.

D. Pollutants from Stack A may disperse more effectively.

Answer: A, B

Why: The inversion layer extends from the surface to 200 m, so pollutants released below 200 m tend to remain trapped near the ground. • Stack A = 80 m: It releases pollutants within/below the inversion layer, so pollutants are more likely to be trapped. Hence, a is correct.

Week 4 — Climate Change

15 questions · 5 MSQ · 10 other

Q1 MCQ — Climate Change

Climate is typically defined based on atmospheric observations averaged over what minimum time period?

A. 1 year

B. 10 years

C. 30 years

D. 100 years.

Answer: C

Why: According to the World Meteorological Organization, climate is commonly described using averages and variability of weather-related parameters over a period of at least 30 years.

Q2 MCQ — Climate Change

Which of the following methods is commonly used to reconstruct past atmospheric temperatures and climate conditions?

A. By relying solely on historical records and written documents

B. By measuring present-day ocean currents

C. By analyzing tiny air bubbles preserved within ice cores

D. By using satellite observations collected since the 1800s

Answer: C

Why: Ice cores extracted from glaciers and ice sheets contain tiny air bubbles that were trapped when the ice formed. These bubbles preserve information about past atmospheric composition (e.g., CO2 and CH4 concentrations), allowing scientists to reconstruct past climate conditions and estimate historical temperatures over thousands of years.

Q3 MCQ — Climate Change

If global photosynthetic activity increased significantly while fossil fuel emissions remained unchanged, how would the atmospheric CO2 record represented by the Keeling Curve most likely respond?

A. A reduced rate of increase in atmospheric CO2 concentration

B. A steeper increase in atmospheric CO2 concentration

C. Complete disappearance of seasonal CO2 fluctuations

D. No measurable change in atmospheric CO2 concentration

Answer: A

Why: Photosynthesis removes CO2 from the atmosphere and stores it in plant biomass. If global photosynthetic activity increased substantially, more CO2 would be absorbed from the atmosphere, partially offsetting fossil fuel emissions. As a result, the Keeling Curve would still show rising atmospheric CO2 concentrations, but the rate of increase would be slower than before.

Q4 MCQ — Climate Change

Which of the following best describes the influence of anthropogenic aerosols on Earth's climate?

A. They contribute only to climate warming.

B. They contribute only to climate cooling.

C. They have no significant effect on climate.

D. They can exert both warming and cooling effects on climate.

Answer: D

Why: Different aerosols affect climate in different ways. Sulfate aerosols generally cool the climate by reflecting incoming solar radiation, whereas black carbon (soot) absorbs sunlight and contributes to warming. Aerosols can also modify cloud properties, leading to additional cooling or warming effects.

Q5 MCQ — Climate Change

One carbon credit represents the offset of 1 tonne of CO2 equivalent (CO2e). A coal-fired power plant emits 2.5 kg of CO2 per kWh of electricity generated and operates continuously at a capacity of 500 MW throughout the year (365 days). How many carbon credits would be required annually to offset 50% of the plant's CO2 emissions? How many would be required to offset 100% of its emissions?

A. 5,475,000 and 10,950,000

B. 4,765,000 and 9,850,000

C. 5,375,000 and 10,450,000

D. 4,565,000 and 9,350,0004

Answer: A

Why: Annual electricity generation = 500 MW 365 days 24 h/day = 4,380,000 MWh = 4.38 109 kWh Annual CO2 emissions = 2.5 kg/kWh 4.38 109 kWh = 1.095 1010 kg = 1.095 107 tons Since 1 carbon credit = 1 tonne CO2e: Credits for 100% offset = 10,950,000 Credits for 50% offset = 10,950,000 / 2 = 5,475,000

Q6 MCQ — Climate Change

Which of the following is a reliable indicator of climate change that is directly linked to rising ocean temperatures?

A. Rise in global average sea levels

B. Ocean acidification

C. Destruction of coral reefs

D. Increased volcanic activity on the ocean floor

Answer: A

Why: Rising ocean temperatures contribute directly to sea level rise through two primary mechanisms: thermal expansion as seawater warms, it expands, increasing sea volume, and melting of glaciers and ice sheets warmer oceans accelerate the melting of ice shelves and glaciers that feed into the sea. As a result, the rise in global average sea levels is a direct and measurable indicator of climate change, closely linked to increasing ocean temperatures.

Q7 MCQ — Climate Change

Which of the following is not generally classified as a geoengineering strategy for mitigating climate change?

A. Ocean fertilization

B. Stratospheric aerosol injection

C. Marine cloud brightening

D. Deep ocean storage of CO2

Answer: D

Why: Geoengineering typically refers to large-scale deliberate interventions in the Earth's climate system, such as ocean fertilization, stratospheric aerosol injection, and marine cloud brightening. These are examples of carbon dioxide removal (CDR) or solar radiation management (SRM) approaches. In contrast, deep-ocean storage of CO2 is generally considered a carbon capture and storage (CCS) technique rather than a geoengineering strategy.

Q8 MCQ — Climate Change

Permafrost thaw and subsidence will directly lead to

A. addition of more greenhouse gases to the atmosphere

B. increased storm surge levels

C. ocean acidification

D. proliferation of invasive species

Answer: A

Why: In the Arctic, large amounts of organic carbon are stored in permafrost. The carbon in the permafrost has built up over thousands of years, as dead plants have been buried and accumulated within layers of frozen soil, where the cold prevents the organic material from decomposing. As the Arctic warms and soils thaw, the organic matter in these soils will decompose rapidly and return to the atmosphere as CO2 or CH4, both important greenhouse gases that further warm the Arctic.

Q9 Assertion–Reason — Climate Change

Assertion (A): The Kyoto Protocol imposed legally binding greenhouse gas (GHG) emission reduction targets on developing countries. Reason (R): Under the principle of common but differentiated responsibilities (CBDR), developing countries were exempted from binding emission reduction targets during the first commitment period of the Kyoto Protocol.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: D

Why: The Kyoto Protocol established binding emission reduction targets for developed (Annex I) countries, not developing countries. The exemption of developing countries was based on the CBDR principle, which recognizes differing historical responsibilities and capacities among nations.

Q10 Match — Climate Change

Match the concepts in List I with their correct descriptions in List II. List I (Concepts) List II (Descriptions)

A. Pre-combustion capture

B. Oxyfuel combustion 1. Fuel is burned in nearly pure oxygen, producing

C. Post-combustion capture a flue gas consisting mainly of CO2 and H2O. 2. A gas separation process that uses liquid solvents to selectively remove CO2. 3. Fuel is converted into synthesis gas (syngas), and CO2 is separated before hydrogen is

D. Absorption combusted. 4. CO2 is removed from flue gas after the fuel has been combusted.

Answer: B

Why: MSQ (5 ? 2 = 10)

Q11 MSQ — select all — Climate Change

Which of the following are major greenhouse gases that occur naturally in Earth's atmosphere?

A. Carbon dioxide (CO2)

B. Methane (CH4)

C. Chlorofluorocarbons (CFCs)

D. Water vapor (H2O)

Answer: A, B, D

Why: Carbon dioxide (CO2), methane (CH4), and water vapor (H2O) are naturally occurring greenhouse gases that help regulate Earth's temperature. Chlorofluorocarbons (CFCs) are synthetic compounds produced by human activities and do not occur naturally in the atmosphere, although they are potent greenhouse gases.

Q12 MSQ — select all — Climate Change

Which of the following statements regarding the role of the ocean in Earth's climate system are correct?

A. The ocean has no significant influence on atmospheric CO2 concentrations.

B. The ocean absorbs approximately one-quarter of anthropogenic CO2 emissions.

C. Thermal expansion of seawater contributes to global sea-level rise.

D. The ocean's large heat capacity and thermal inertia can delay surface warming over long timescales.

Answer: B, C, D

Why: The ocean is a major regulator of Earth's climate. It absorbs roughly 25% of human-generated CO2 emissions, helping to moderate atmospheric CO2 levels. As ocean water warms, it expands, contributing to sea-level rise.

Q13 MSQ — select all — Climate Change

Which of the following factors can influence Earth's energy budget?

- **A.** Land-use changes
- **B.** Anthropogenic aerosols
- **C.** The extent of sea ice and seasonal snow cover
- **D.** The magnitude of ocean heat uptake

Answer: A, B, C, D

Why: All four factors affect Earth's energy balance. Land-use changes alter surface albedo and energy fluxes. Anthropogenic aerosols influence the absorption and reflection of solar radiation and can modify cloud properties.

Q14 MSQ — select all — Climate Change

Which of the following are examples of cascading climate impacts?

- **A.** Melting ice reduces Earth's albedo, leading to additional warming and further ice loss.
- **B.** Sea-level rise increases the risk of storm-surge flooding, causing infrastructure damage and economic losses.
- **C.** Rising temperatures reduce crop yields, leading to higher food prices and adverse health outcomes.
- **D.** An isolated impact that remains confined to a single sector without triggering secondary effects.

Answer: A, B, C

Why: Cascading climate impacts occur when an initial climate-related change triggers a chain of interconnected consequences across environmental, social, or economic systems. Options a, b, and c illustrate such cascading effects. Option d is not a cascading impact because it involves no secondary or downstream consequences.

Q15 MSQ — select all — Climate Change

Which of the following are primary functions of the IPCC?

- **A.** Reviewing and synthesizing the latest climate science data from peer-reviewed literature
- **B.** Conducting original climate research and experiments
- **C.** Assessing the socioeconomic impacts of climate change
- **D.** Producing reports that inform global policy on climate change adaptation and mitigation

Answer: A, C, D

Why: The IPCC primarily reviews existing scientific literature (a), assesses climate change impacts, including socioeconomic aspects (c), and produces reports to guide global climate policy (d). It does not conduct original research or experiments (b).

Week 5 — Renewable Energy I

15 questions · 5 MSQ · 10 other

Q1 MCQ — Renewable Energy I

A country decides to rapidly phase out fossil fuels without investing adequately in alternative energy infrastructure. Which dimension of the Global Energy Trilemma is most likely to be adversely affected in the short term? I. Energy security II. Energy equity III. Environmental sustainability

- **A.** I only
- **B.** I and II only
- **C.** II and III only
- **D.** I, II, and III

Answer: A

Why: The most immediate and direct consequence is a reduction in the reliability and resilience of energy supply, which affects energy security. Effects on energy equity are possible but not certain from the information provided, while environmental sustainability would generally improve.

Q2 MCQ — Renewable Energy I

Which of the following statements about a Trombe wall are correct? I. It is a passive solar heating system. II. It stores solar energy in a high thermal-mass wall. III. It requires photovoltaic panels to generate electricity. IV. It can help reduce space-heating demand in buildings.

- **A.** I and II only
- **B.** I, II, and IV only
- **C.** III and IV only
- **D.** I, II, III, and IV

Answer: B

Why: A Trombe wall is a passive solar design feature consisting of a sun-facing glazed wall with high thermal mass that absorbs, stores, and gradually releases heat to the building interior. It does not require photovoltaic panels and is intended for thermal energy management rather than electricity generation.

Q3 Assertion–Reason — Renewable Energy I

Assertion (A): Bifacial photovoltaic (PV) modules can generate more electricity than monofacial PV modules under suitable operating conditions. Reason (R): Bifacial PV modules are capable of capturing solar radiation from both their front and rear surfaces.

- **A.** Both A and R are true, and R is the correct explanation of A
- **B.** Both A and R are true but R is not the correct explanation of A
- **C.** A is true but R is false
- **D.** A is false but R is true

Answer: A

Why: The assertion is true because bifacial PV modules can produce more electricity than conventional monofacial modules when installed under favorable conditions. The reason is also true, as bifacial modules are designed to capture solar radiation on both sides of the panel. In addition to direct sunlight incident on the front surface, the rear surface can utilize reflected and diffuse radiation from the ground and surrounding surfaces (albedo radiation).

Q4 MCQ — Renewable Energy I

Which of the following concentrating solar power (CSP) technologies is the most commercially mature and accounts for the largest share of installed CSP capacity worldwide?

A. Solar towers

B. Parabolic dishes

C. Parabolic troughs

D. Linear-Fresnel reflectors

Answer: C

Why: Parabolic trough systems are the most established CSP technology, with extensive commercial deployment and operational experience across many large-scale solar thermal power plants worldwide.

Q5 MCQ — Renewable Energy I

What is the primary purpose of thermal energy storage in a concentrating solar power (CSP) plant?

A. To store excess thermal energy for electricity generation during periods of low or no solar radiation

B. To increase the concentration ratio of solar collectors

C. To convert solar energy directly into electricity without a turbine

D. To eliminate heat losses from the solar receiver

Answer: A

Why: Thermal energy storage systems, commonly based on molten salts, store heat collected during sunny periods and release it later to generate electricity when sunlight is unavailable, thereby improving the dispatchability of CSP plants.

Q6 MCQ — Renewable Energy I

According to Betz's law, a wind turbine cannot extract all of the kinetic energy from the wind because

A. the air must retain some kinetic energy after passing through the rotor.

B. the wind speed must remain unchanged across the rotor plane.

C. the air accelerates as it passes through the rotor, reducing energy extraction.

D. the kinetic energy of the wind is independent of its velocity.

Answer: A

Why: A wind turbine generates power by extracting a portion of the wind's kinetic energy and reducing its velocity. However, the air must retain some kinetic energy after passing through the rotor so that it can continue flowing downstream. If all of the kinetic energy were extracted, the air would come to a complete stop behind the turbine, preventing any further airflow through the rotor and making continuous power generation impossible.

Q7 MCQ — Renewable Energy I

Which of the following is a primary advantage of offshore wind farms compared with onshore wind farms?

A. They eliminate the need for transmission infrastructure.

B. They generally experience stronger and more consistent wind resources.

C. They require smaller turbines to achieve the same power output.

D. They are less expensive to install and maintain.

Answer: B

Why: Offshore locations typically experience higher average wind speeds and lower turbulence than land-based sites. These conditions increase energy production and improve the capacity factor of offshore wind turbines, making offshore wind farms an attractive option despite their higher installation and maintenance costs.

Q8 MCQ — Renewable Energy I

A passive solar building receives 5,000 kWh of useful solar heat through its south-facing windows during the winter. If the building experiences a total heat loss of 4,000 kWh over the same period, what is the additional heating energy required from a backup heating system?

A. 0 kWh

B. 1,000 kWh

C. 4,000 kWh

D. 5,000 kWh

Answer: A

Why: The solar heat gain (5,000 kWh) exceeds the building's heat loss (4,000 kWh). Since the available solar heat is sufficient to offset all heat losses, no additional heating is required from a backup system. The excess 1,000 kWh represents a net heat surplus.

Q9 MCQ — Renewable Energy I

For a fixed solar photovoltaic (PV) system in the Northern Hemisphere, increasing the panel tilt angle beyond the annual optimum generally helps to

A. increase winter energy capture when the sun is lower in the sky

B. increase summer energy capture when the sun is nearly overhead

C. eliminate reflection losses from the panel surface

D. maintain the same solar incidence angle throughout the year

Answer: A

Why: In winter, the sun follows a lower path across the sky. A steeper panel tilt improves the angle at which sunlight strikes the module, thereby increasing winter solar energy capture.

Q10 MCQ — Renewable Energy I

Which feature most clearly distinguishes an active solar heating system from a passive solar heating system?

A. Collection of solar energy through south-facing glazing

B. Storage of heat in materials with high thermal mass

C. Distribution of heat by natural convection and radiation

D. Heat transfer assisted by pumps or fans

Answer: D

Why: Active solar heating systems use mechanical devices such as pumps or fans to circulate a heat-transfer fluid and distribute solar heat. Passive solar heating systems rely on building design, thermal mass, and natural heat transfer mechanisms such as conduction, convection, and radiation. MSQ (5 ◆ 2 = 10)

Q11 MSQ — select all — Renewable Energy I

Which of the following factors are typically considered when estimating the technical potential of solar energy in a region?

A. Available land or rooftop area

B. Solar resource availability

C. Module conversion efficiency

D. Electricity market prices

Answer: A, B, C

Why: Technical potential accounts for resource availability, suitable installation area, and technology performance. Electricity prices are relevant to economic potential.

Q12 MSQ — select all — Renewable Energy I

Which of the following are potential environmental concerns associated with wind energy projects?

A. Bird and bat mortality due to turbine collisions

B. Habitat fragmentation caused by access roads and supporting infrastructure

C. Noise and shadow-flicker effects on nearby communities

D. Alteration of local landscape aesthetics and visual character

Answer: A, B, C, D

Why: Although wind energy produces electricity with minimal direct emissions, wind farms may raise environmental concerns related to wildlife impacts, habitat disturbance, noise and shadow flicker, and visual effects on the surrounding landscape.

Q13 MSQ — select all

Renewable Energy I

Which of the following pathways can be used to produce green hydrogen directly using solar energy?

- **A.** Photoelectrochemical water splitting
- **B.** Photocatalytic water splitting
- **C.** Photovoltaic-powered water electrolysis
- **D.** Solar-powered alkaline electrolysis

Answer: A, B

Why: Photoelectrochemical and photocatalytic systems use sunlight directly to drive water-splitting reactions. In contrast, photovoltaic-powered and solar-powered electrolysis first convert solar energy into electricity, which is then used to produce hydrogen.

Q14 MSQ — select all

Renewable Energy I

Which of the following factors can influence the electrical output of a photovoltaic (PV) module?

- **A.** Solar irradiance incident on the module surface
- **B.** Module operating temperature
- **C.** Shading of individual cells within the module
- **D.** Orientation and tilt angle of the module

Answer: A, B, C, D

Why: PV output is influenced by the amount of sunlight received, module temperature, shading effects, and installation geometry.

Q15 MSQ — select all

Renewable Energy I

Which of the following represents a potential environmental challenge associated with the large-scale deployment of renewable energy technologies?

- **A.** Unintended lifecycle carbon footprints from the manufacturing and disposal of renewable energy components
- **B.** Ecological fragmentation and shifts in biodiversity due to large-scale renewable infrastructure
- **C.** Accelerated resource depletion and environmental degradation linked to the mining of critical raw materials for green technologies
- **D.** Increased reliance on geologically constrained and environmentally disruptive energy storage solutions

Answer: A, B, C

Why: Large-scale renewable energy technologies can pose environmental challenges, such as carbon footprints from manufacturing and disposal, ecological disruption affecting biodiversity, and resource depletion due to the mining of critical materials needed for these technologies.

Week 6 — Renewable Energy II

15 questions · 5 MSQ · 10 other

Q1 MCQ — Renewable Energy II

Which of the following geothermal energy technologies involves creating engineered underground reservoirs by enhancing or artificially stimulating rock formations, thereby enabling the extraction of geothermal energy from locations that would otherwise be inaccessible?

A. Dry steam power plants

B. Flash steam power plants

C. Binary cycle power plants

D. Enhanced geothermal systems

Answer: D

Why: Enhanced geothermal systems create artificial underground reservoirs by injecting fluid into hot, impermeable rock formations to improve permeability and heat extraction. This allows geothermal energy to be harnessed from regions where natural geothermal reservoirs are absent or inaccessible.

Q2 MCQ — Renewable Energy II

A tidal barrage power plant has a basin area of 470 km2 and operates with a tidal range of 8 m. The tidal cycle period is 12 h. Assuming a seawater density of 1000 kg m3 and gravitational acceleration of 9.81 m s2, what is the approximate average power output of the plant?

A. 3.4 GW

B. 34 GW

C. 340 GW

D. 0.34 GW

Answer: A

Why: The average power (Pavg) harvested from a tidal basin is (.g.A.h2) / (2T) Given: = 1000 kg m3, g = 9.81 m s2, A = 470 $\times$ 106 m2, h = 8 m, T = 12 $\times$ 3600 s Substituting, Pavg = (1000 $\times$ 9.81 $\times$ 470 $\times$ 106 $\times$ 82) / (2 $\times$ 12 $\times$ 3600) = 3.42 $\times$ 109 W = 3.4 GW

Q3 MCQ — Renewable Energy II

Which of the following ocean energy technologies has the largest estimated global technical energy potential?

A. Tidal energy

B. Wave energy

C. Ocean thermal energy conversion

D. Salinity gradient energy

Answer: C

Why: OTEC exploits the temperature difference between warm surface water and cold deep ocean water. Because tropical oceans cover vast areas and operate continuously, OTEC has the largest estimated global technical potential, often exceeding 44,000 TWh per year, which is significantly greater than the estimated potentials of tidal, wave, or salinity-gradient energy technologies.

Q4 Assertion–Reason — Renewable Energy II

Assertion (A): Fast pyrolysis produces bio-oil by rapidly heating biomass in the absence of oxygen. Reason (R): The absence of oxygen prevents combustion and enables the thermal decomposition of biomass into liquid products.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A

Why: Fast pyrolysis involves heating biomass rapidly in an oxygen-free environment. The absence of oxygen prevents the biomass from burning and instead promotes thermal decomposition, producing bio-oil as the major product along with biochar and syngas. Therefore, the reason correctly explains the assertion.

Q5 MCQ — Renewable Energy II

Point absorbers and attenuators are classified as oscillating-body wave energy converters because they primarily

A. convert pressure waves propagating through a flexible, water-filled tube into electrical energy.

B. convert the oscillatory motion of floating or submerged structures induced by ocean waves into useful power.

C. generate electricity from bidirectional airflow produced by the oscillation of a water column.

D. harness the potential energy of seawater collected in an elevated reservoir by wave overtopping.

Answer: B

Why: Point absorbers and attenuators generate electricity by converting the oscillatory motion of floating or submerged structures induced by ocean waves into useful power. This wave-induced body motion is the defining feature of oscillating-body wave energy converters.

Q6 MCQ — Renewable Energy II

A hydroelectric power plant operates at a head of 55 m with a water flow rate of 500 m3 s^{-1} through the turbines. If the overall efficiency of the plant is 77%, calculate the electricity generation capacity. Assume water density = 1000 kg m^{-1} and g = 9.81 m s-2.

A. 207 MW

B. 207 kW

C. 207 GW

D. 207 W

Answer: A

Why: Mass flow of water (mw) = 500 m3 s^{-1} $\times$ 1000 kg m^{-3} = 500,000 kg s^{-1} Therefore, Pelec = ehydromwg(z) = 0.77 $\times$ 500,000 $\times$ 9.81 $\times$ 55 = 2.07 $\times$ 108 W = 207 MW

Q7 MCQ — Renewable Energy II

In the anaerobic digestion of food waste, which stage involves the breakdown of complex organic polymers such as carbohydrates, proteins, and lipids into simpler soluble compounds that can be utilized by microorganisms?

A. Acidogenesis

B. Acetogenesis

C. Hydrolysis

D. Methanogenesis

Answer: C

Why: Hydrolysis is the first stage of anaerobic digestion, during which complex organic matter is converted into simpler soluble molecules such as sugars, amino acids, and fatty acids, making them available for subsequent microbial degradation.

Q8 MCQ — Renewable Energy II

Which of the following best distinguishes third-generation biofuels from first- and second- generation biofuels?

A. They are produced exclusively from food crops such as corn and sugarcane.

B. They are derived from lignocellulosic biomass such as agricultural residues and woody biomass.

C. They are produced primarily from algae and other high-lipid microorganisms.

D. They are synthesized directly from fossil fuels blended with ethanol.

Answer: C

Why: Third-generation biofuels are mainly derived from algae and other microorganisms, which can achieve high biomass productivity without competing directly with food crops.

Q9 MCQ — Renewable Energy II

Wave energy is often described as an indirect form of solar energy because

A. Solar heating of the Earth's surface generates winds that transfer energy to ocean waves.

B. Solar radiation directly heats ocean water, producing surface waves.

C. Wave motion is primarily caused by the thermal expansion of seawater.

D. Ocean waves are generated by tidal forces induced by solar radiation.

Answer: A

Why: Most ocean waves are generated by wind. Since winds are driven by the uneven heating of the Earth's surface by the Sun, wave energy is considered an indirect form of solar energy.

Q10 MCQ — Renewable Energy II

Which of the following characteristics distinguishes geothermal energy from many other renewable energy resources?

A. It is available only during daylight hours.

B. It can provide reliable, continuous baseload power.

C. Its output is directly governed by seasonal weather conditions.

D. It relies on large-scale energy storage to ensure a stable power supply.

Answer: B

Why: Unlike many renewable energy sources that are influenced by weather conditions, geothermal energy can provide a continuous and reliable supply of power, making it well-suited for baseload electricity generation. MSQ (5 × 2 = 10)

Q11 MSQ — select all — Renewable Energy II

Which of the following are potential benefits of biomass pretreatment intended for biofuel production?

A. Increasing the surface area available for enzymatic hydrolysis.

B. Reducing cellulose crystallinity and improving enzyme accessibility.

C. Disrupting the lignin–carbohydrate matrix to expose fermentable polysaccharides.

D. Directly converting cellulose and hemicellulose into biofuels during the pretreatment step.

Answer: A, B, C

Why: The primary goals of biomass pretreatment are to increase enzyme accessibility by modifying the physical and chemical structure of lignocellulosic biomass. Pretreatment facilitates subsequent hydrolysis and fermentation but does not directly convert biomass into biofuels.

Q12 MSQ — select all — Renewable Energy II

Which of the following environmental impacts are commonly associated with the construction and operation of large hydroelectric dams?

A. Alteration of natural river flow patterns.

B. Disruption of fish migration and aquatic ecosystems.

C. Submergence of terrestrial habitats due to reservoir formation.

D. Emission of fly ash during electricity generation.

Answer: A, B, C

Why: Large hydroelectric dams can modify natural river flow regimes, impede fish migration, and inundate terrestrial ecosystems through reservoir creation. Fly ash emissions are associated with coal-fired power plants and do not occur during hydroelectric power generation.

Q13 MSQ — select all

Renewable Energy II

Which of the following statements about tidal energy are correct?

- **A.** It is driven primarily by the gravitational interaction of the Earth, Moon, and Sun.
- **B.** It is one of the most predictable renewable energy resources.
- **C.** Tidal barrages utilize differences in water level across a basin.
- **D.** The magnitude of tidal currents and tidal ranges is largely independent of local coastal geography.

Answer: A, B, C

Why: Tidal energy is governed by gravitational forces and is highly predictable. Tidal barrages exploit differences in water levels, while local coastal geometry strongly influences tidal ranges and current velocities.

Q14 MSQ — select all

Renewable Energy II

Which of the following are common direct-use applications of geothermal energy (i.e., where geothermal heat is utilized without first being converted into electricity?

- **A.** Fueling public transportation systems.
- **B.** District heating of residential and commercial buildings.
- **C.** Heating of greenhouses for year-round crop production.
- **D.** Aquaculture and fish farming operations.

Answer: B, C, D

Why: Geothermal heat can be used directly for space heating, greenhouse heating, and aquaculture. Transportation systems are generally powered by liquid fuels, electricity, or hydrogen rather than direct geothermal heat.

Q15 MSQ — select all

Renewable Energy II

Which of the following are recognized environmental benefits associated with the production and use of biodiesel?

- **A.** Reduced reliance on petroleum-based fuels.
- **B.** Complete elimination of exhaust emissions from diesel engines.
- **C.** Reduced environmental persistence in the event of accidental spills.
- **D.** Valorization of waste oils and animal fats as feedstocks.

Answer: A, C, D

Why: Biodiesel can reduce dependence on petroleum fuels, is generally more biodegradable in the event of accidental spills, and can be produced from waste oils and animal fats. However, biodiesel combustion still generates exhaust emissions and does not eliminate them completely.

Week 7 — Circular Economy

10 questions · 0 MSQ · 10 other

Q1 MCQ — Circular Economy

Which of the following activities is not an example of reuse?

A. Refilling a beverage bottle for repeated use

B. Using a refillable shopping bag multiple times

C. Melting and reprocessing plastic bottles to manufacture new bottles

D. Donating used clothing for continued use by others

Answer: C

Why: Reuse involves using a product again with little or no modification, whereas recycling involves processing waste materials to create new products. Option (c) is an example of recycling, not reuse.

Q2 MCQ — Circular Economy

According to the circular economy hierarchy, which of the following sequences correctly arranges the listed "R" strategies in increasing order of circularity (from least circular to most circular)?

A. Recycle < Repurpose < Rethink < Repair

B. Repurpose < Recycle < Repair < Rethink

C. Recycle < Repurpose < Repair < Rethink

D. Recycle < Repair < Repurpose < Rethink

Answer: C

Why: Recycling recovers materials after a product reaches the end of its useful life and is generally less circular than strategies that retain product value. Repurposing gives a product a new function, while repairing extends its original use with minimal resource input. Rethinking involves redesigning products, services, or consumption patterns to avoid waste generation in the first place and is therefore considered the most circular strategy among the options.

Q3 Assertion–Reason — Circular Economy

Assertion (A): Industrial symbiosis can improve resource efficiency within an industrial cluster. Reason (R): The waste or by-products generated by one industry can be used as inputs by another industry.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A

Why: Industrial symbiosis enhances resource efficiency by facilitating the exchange of materials, energy, and by-products among industries. The reason directly explains the assertion.

Q4 MCQ — Circular Economy

A company decides to remanufacture electric motors instead of producing new ones. A life-cycle assessment shows that manufacturing a new motor results in 5.0 tonnes of CO2 emissions, whereas remanufacturing an existing motor results in 1.5 tonnes of CO2 emissions. If the company remanufactures 65 motors, how much CO2 emissions are avoided compared with manufacturing the same number of new motors?

A. 201.5 tonnes CO2

B. 227.5 tonnes CO2

C. 240.5 tonnes CO2

D. 280.4 tonnes CO2

Answer: B

Why: Emission savings per motor: 5.0 – 1.5 = 3.5 tonnes CO2 For 65 motors: 65 – 3.5 = 227.5 tonnes CO2 Remanufacturing each motor avoids 3.5 tonnes of CO2 emissions compared with manufacturing a new motor. Therefore, remanufacturing 65 motors results in a total emissions savings of 227.5 tonnes of CO2.

Q5 MCQ — Circular Economy

Which of the following best illustrates a "rethink" strategy in a circular economy, where a product or system is fundamentally redesigned to deliver greater value and resource efficiency?

A. Designing beverage containers that can be returned, cleaned, and refilled multiple times.

B. Recovering phosphorus from municipal wastewater for fertilizer production.

C. Replacing individual office printers with a centralized on-demand printing service.

D. Repairing industrial motors to extend their operational lifespan.

Answer: C

Why: A rethink strategy involves fundamentally changing how a product or service is delivered to reduce resource consumption and improve utilization. Replacing individual printers with a shared on-demand printing service rethinks the conventional ownership model. In contrast, options (a), (b), and (d) represent reuse, resource recovery, and repair, respectively.

Q6 MCQ — Circular Economy

Within the framework of a circular economy, what is the most significant advantage of repurposing or recycling decommissioned wind turbine blades?

A. It reduces the need for landfill disposal while retaining the value embedded in composite materials.

B. It decreases the environmental impacts associated with the extraction and processing of virgin construction materials.

C. It creates opportunities for secondary applications such as bridges, noise barriers, and architectural structures.

D. It extends the usefulness of blade materials beyond their original function, thereby slowing resource throughput in the economy.

Answer: D

Why: All four statements are valid benefits. However, from a circular economy perspective, the most fundamental objective is to retain the value of materials for as long as possible and reduce the flow of virgin resources through the economy. Repurposing wind turbine blades directly achieves this by extending the useful life of embedded materials.

Q7 MCQ — Circular Economy

Two cities consume the same quantity of energy and materials each year. City A recovers and reuses a substantial fraction of its waste streams, whereas City B primarily relies on landfill disposal. Which statement best describes the likely difference in their urban metabolisms?

A. City A is likely to have a lower material throughput per unit of useful service delivered.

B. City A is likely to exhibit higher material circulation within the system and lower dependence on virgin resource inputs.

C. City A is likely to accumulate waste stocks more rapidly because recovered materials remain within the city.

D. City A is likely to have a larger metabolic footprint because resource recovery processes require additional energy and infrastructure.

Answer: B

Why: By recovering and reusing a larger fraction of its waste streams, City A keeps materials in productive use for longer periods, increasing material circulation within the urban system. This reduces the need for virgin resource extraction and reflects a more circular and resource-efficient urban metabolism compared with City B, which primarily relies on landfill disposal.

Q8 Match — Circular Economy

Match the following circular business models with their most appropriate defining characteristics. Circular Business Model Key Characteristic

A. Circular supplies 1. Providing access to a product through leasing or subscription rather than ownership

B. Product as a service 2. Using renewable, recyclable, or sustainably sourced inputs

C. Product life extension 3. Extending product usability through repair, refurbishment, or remanufacturing

D. Sharing platform 4. Enabling the shared use of underutilized products or assets

Answer: A

Why: Circular supplies rely on renewable or recyclable inputs, while product-as-a-service models provide access to products without transferring ownership. Product life extension focuses on repair and remanufacturing, whereas sharing platforms increase the utilization of underused assets through collaborative access.

Q9 MCQ — Circular Economy

Two companies adopt different approaches for managing end-of-life office chairs. One company repairs damaged chairs and returns them to service, while the other recycles the chairs by processing them into raw materials for new products. According to the Butterfly Diagram of the circular economy, why is the first approach generally considered more circular?

A. Repair requires less labor and economic activity than recycling.

B. Repair preserves a greater proportion of the product's embodied value than material recycling.

C. Repair completely eliminates the need for future resource extraction.

D. Repair prevents the generation of waste throughout the product life cycle.

Answer: B

Why: The Butterfly Diagram prioritizes strategies that retain the highest value of products, components, and materials. By keeping the chair in use with minimal processing, repair preserves more embodied value than recycling, which breaks the product down to its constituent materials.

Q10 MCQ

Circular Economy

A city achieves a recycling rate of 95% for aluminum cans. Which statement best reflects a key limitation of relying solely on circular economy strategies to achieve long-term resource sustainability?

A. High recycling rates can substantially reduce demand for primary aluminum, but may not eliminate it because material losses occur and overall consumption can continue to grow.

B. Once recycling rates exceed a critical threshold, the aluminum cycle becomes effectively self-sustaining and independent of virgin resource inputs.

C. High recycling rates ensure that the environmental impacts associated with aluminum production are largely eliminated over time.

D. Increasing recycling efficiency is sufficient to decouple economic growth from resource extraction under all circumstances.

Answer: A

Why: Even highly efficient recycling systems experience material losses, and increasing consumption can outpace gains from resource recovery. Therefore, high recycling rates alone do not guarantee complete circularity or eliminate the need for virgin resource extraction. .

Week 8 — Water

20 questions · 0 MSQ · 20 other

Q1 MCQ — Water

Consider the following statements regarding water resources and water use: I. Groundwater constitutes a larger share of freshwater resources than all fresh surface water combined. II. Since groundwater forms nearly one-third of freshwater resources, it is generally more accessible than surface water. III. The amount of usable freshwater on Earth is less than 0.1% of the planet's total water resources. IV. Most freshwater surface water is stored in rivers. Which of the above statements are correct?

A. I and II only

B. I and III only

C. II and IV only

D. I, III and IV only

Answer: B

Why: I. Correct. Groundwater constitutes about 30.1% of freshwater resources, whereas surface water accounts for only about 0.3%.

Q2 MCQ — Water

Consider the following statements regarding water withdrawal and consumption: I. Water used for drinking and cooking is generally considered a return flow. II. Municipal wastewater discharge can exceed municipal water supply. III. Water used for bathing generally contributes to wastewater generation. IV. Domestic water use in urban areas is predominantly non-consumptive in nature. Which of the above statements are correct?

A. I and II only

B. I, II and III only

C. II and III only

D. II, III and IV only

Answer: D

Why: I. Incorrect. Water used for drinking and cooking is a consumptive use and is generally not returned directly to the water source.

Q3 MCQ — Water

Consider the following pollution sources: I. Industrial effluent discharged through a pipeline. II. Stormwater runoff from urban roads. III. Agricultural runoff carrying fertilizers and pesticides. IV. Wastewater treatment plant discharge. V. Leachate entering a river through a distinct and identifiable channel. How many of the above are point sources of pollution?

A. 2

B. 3

C. 4

D. 5

Answer: B

Why: Point sources are pollution sources that originate from a specific, identifiable location such as a pipe, drain, or outlet. Source I is a point source because industrial effluent is discharged through a clearly identifiable pipeline. Source II is a non-point source.

Q4 MCQ Water

The concentrations of various ions present in a groundwater sample are provided in the table below. Using the given data, calculate the ionic strength of the water sample in milli-moles/L.

A. 3.25 mM

B. 6.50 mM

C. 2.13 mM

D. 4.38 mM

Answer: A

Why: From assignment detailed solution.

Q5 MCQ Water

At a certain temperature, pKw = 13.4. The hydrogen ion concentration of a water sample is 10-5.4 M. The nature of the sample is:

A. Acidic

B. Alkaline

C. Neutral

D. Cannot be determined

Answer: A

Why: Neutral pH: pH(neutral)= pKw/2 = 13.4/2 = 6.7 Given: pH = 5.4 Since, 5.4 < 6.7, the sample is acidic.

Q6 MCQ Water

A 500 mL water sample was evaporated to dryness in a pre-weighed dish and then ignited in a muffle furnace. The following observations were recorded: i. Mass of empty dish = 52.615 g ii. Mass of dish and residue after drying at 105°C = 52.965 g iii. Mass of dish and residue after ignition at 550°C = 52.815 g Which one of the following is correct?

A. TS = 700 mg/L, TVS = 400 mg/L, TFS = 300 mg/L

B. TS = 700 mg/L, TVS = 300 mg/L, TFS = 400 mg/L

C. TS = 400 mg/L, TVS = 300 mg/L, TFS = 100 mg/L

D. TS = 350 mg/L, TVS = 150 mg/L, TFS = 200 mg/L

Answer: B

Why: Mass of residue after drying: 52.965 - 52.615 = 0.350 g= 350 mg 0.5 L Sample volume: 500 mL = 700 mg/L TS = 350 mg / 0.5 L = Total Fixed Solids (TFS): Mass remaining after ignition: 52.815 - 52.615 = 0.200 g = 200 mg TFS = 200 mg / 0.5 L = 400 mg/L Total Volatile Solids (TVS): TVS = TS - TFS = 700 - 400 = 300 mg/L

Q7 MCQ Water

A groundwater sample was analyzed and found to contain 64 mg/L of calcium (Ca^{2+}), 21.9 mg/L of magnesium (Mg^{2+}), 57.5 mg/L of sodium (Na), 31.3 mg/L of potassium (K), 42.6 mg/L of chloride (Cl) and 90 mg/L of carbonate (CO_3^{2-}). Determine the Total Hardness (TH), Carbonate Hardness (CH) and Non- carbonate Hardness (NCH) of the sample.

A. TH = 250 mg/L as CaCO, CH = 250 mg/L as CaCO, NCH = 0 mg/L as CaCO

B. TH = 150 mg/L as CaCO, CH = 250 mg/L as CaCO, NCH = 100 mg/L as CaCO

C. TH = 250 mg/L as CaCO, CH = 150 mg/L as CaCO, NCH = 100 mg/L as CaCO

D. TH = 250 mg/L as CaCO, CH = 100 mg/L as CaCO, NCH = 150 mg/L as CaCO

Answer: C

Why: Equivalent weight of CaCO = Molecular weight / Valency = 100 / 2 = 50 Equivalent weight of 2 Equivalent weight Ca^{2+} = 40 / = 20 Equivalent weight of 2 of Mg^{2+} = 24.3 / 2 = 12.15 CO_3^{2-} = 60 / = 30 Total hardness = [Ca^{2+}] * (50 / 20) + [Mg^{2+}] * (50 / 12.15) = 64 * (50 / 20) + 21.9 * (50 / 12.15) = 160 + 90 = 250 mg/L as CaCO Carbonate hardness = [CO_3^{2-}] * (50 / 30) = 90 * (50 / 30) = 150 mg/L as CaCO Non-carbonate hardness = Total hardness - Carbonate hardness = 250 - 150 = 100 mg/L as CaCO

Q8 MCQ Water

Consider the following organic compounds- methane (CH), methanol (CHOH), formaldehyde (CHO) and carbon dioxide (CO). Arrange them in decreasing order of their theoretical oxygen demand (ThOD) per unit mass.

A. CH > CHO > CHOH > CO

B. CHO > CHOH > CH > CO

C. CHOH > CH > CHO > CO

D. CH > CHOH > CHO > CO

Answer: D

Why: For complete oxidation: Methane: CH + 2O CO + 2HO Molecular CH = mg/mmol O weight of = 16 ThOD 16 = 4.0 mg required 64 CH = 64 / mg O/mg Methanol: CHOH + 1.5O CO + 2HO mg/mmol Molecular weight of CHOH = 32 mg O required = 48 CHOH ThOD = 48 / 32 = 1.5 mg O/mg Formaldehyde: CHO + O CO + HO CHO mg/mmol Molecular weight of = 30 = mg O required = 32 CHO ThOD = 32 / 30 1.07 mg O/mg Carbon dioxide: CO is already completely oxidized. ThOD = 0 Hence, CH > CHOH > CHO > CO

Q9 MCQ Water

Two wastewater samples, A and B, have the same ultimate biochemical oxygen demand. The 5-day BOD of sample A at 20°C is observed to be twice the 5-day BOD of sample B at the same temperature. If the deoxygenation rate constant of sample A is 0.40 day^{-1}, determine the deoxygenation rate constant of sample

B. a. 0.081 day^{-1}

C. 0.146 day^{-1}

D. 0.200 day^{-1}

Answer: B

Why: For sample A, BOD(A) = L * (1 - e^(-0.40 * 5)) = L * (1 - e^(-2)) = L * (1 - 0.1353) = 0.8647L Given, BOD(A) = 2 * BOD(B) Therefore, BOD(B) = 0.8647L / 2 = 0.43235L B, For sample e^(-5k)) BOD(B) = L * (1 - e^(-5k)) e^(-5k) 0.43235L = L * (1 - 0.56765 ln(0.56765) 0.43235 = 1 - e^(-5k) = -5k = k = -ln(0.56765) / 5 = 0.113 day^{-1}

Q10 MCQ Water

A wastewater sample was tested using two BOD bottles. The first bottle contained 10 mL of sample diluted to 300 mL, while the second bottle contained 20 mL of sample diluted to 300 mL. The measured DO depletions after 5 days were 4.2 mg/L and 8.4 mg/L, respectively. Neglecting blank correction. The ratio of the BOD values calculated from the two bottles is:

A. 2

B. 1.5

C. 1

D. 0.5

Answer: C

Why: For Bottle 1, 0.0333 P = 10 / 300 = 126 mg/L BOD = DO depletion / P = 4.2 / 0.0333 = For Bottle 2, P = 20 / 300 = 0.0667 BOD = DO depletion / P = 8.4 / 0.0666 = 126 mg/L Ratio of BOD values = BOD (Bottle 1) / BOD (Bottle 2) = 126 / 126 = 1

Q11 MCQ Water

A wastewater sample has a total ultimate biochemical oxygen demand (Total BOD) of 520 mg/L. The sample contains 35 mg/L of ammonia nitrogen (NH-N). The theoretical nitrogenous biochemical oxygen demand (NBOD) and carbonaceous biochemical oxygen demand (CBOD) of the sample, respectively, are:

A. NBOD = 160 mg/L, CBOD = 360 mg/L

B. NBOD = 140 mg/L, CBOD = 380 mg/L

C. NBOD = 120 mg/L, CBOD = 400 mg/L

D. NBOD = 100 mg/L, CBOD = 420 mg/L

Answer: A

Why: Theoretical NBOD = [NH-N] * 4.57 = 35 * 4.57 = 159.95 mg/L CBOD + mg/L 160 NBOD Since, TBOD = CBOD = 520 - 160 = 360 mg/L

Q12 Assertion–Reason Water

Assertion (A): A wastewater sample having a higher ThOD than another sample may exhibit a lower BOD. Reason (R): ThOD represents the total oxygen required for complete chemical oxidation, whereas BOD depends on the biodegradable fraction oxidized within five days. Choose the correct option:

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A

Why: ThOD represents the theoretical oxygen required for complete oxidation of a compound. BOD represents the oxygen consumed due to biological oxidation during the first five days. A sample may have a high ThOD but contain compounds that are not readily biodegradable.

Q13 MCQ Water

In a river reach, the dissolved oxygen (DO) saturation concentration is measured to be 8.8 mg/L. Using the Streeter–Phelps oxygen sag concept, the dissolved oxygen deficit at a particular location downstream is estimated as 3.6 mg/L. The actual dissolved oxygen concentration at that location is

A. 3.6 mg/L

B. 5.2 mg/L

C. 8.8 mg/L

D. 12.4 mg/L

Answer: B

Why: Dissolved oxygen deficit = DO saturation concentration - Actual DO concentration = 8.8 mg/L - DO(actual) 3.6 mg/L DO(actual) = 5.2 mg/L

Q14 MCQ Water

A river carrying a flow of 1740 m³/s has an upstream biochemical oxygen demand (BOD) concentration of 14.8 mg/L and a dissolved oxygen (DO) concentration of 5.6 mg/L. The river receives wastewater discharges from two outfalls situated on opposite banks. The outfall on the left bank discharges 28 m³/s of wastewater having a BOD concentration of 248 mg/L and a DO concentration of 1.2 mg/L. The outfall on the right bank discharges 32 m³/s of wastewater, but both its BOD and DO concentrations are unknown. Assuming complete and instantaneous mixing of the river water and the wastewater streams, the BOD and DO concentrations measured immediately downstream of the confluence are found to be 21.6 mg/L and 5.45 mg/L, respectively. The BOD and DO concentrations of the wastewater discharged from the right-bank outfall are, respectively:

A. BOD = 183.3 mg/L, DO = 1.0 mg/L

B. BOD = 193.3 mg/L, DO = 0.8 mg/L

C. BOD = 203.3 mg/L, DO = 1.2 mg/L

D. BOD = 193.3 mg/L, DO = 1.0 mg/L

Answer: D

Why: Total downstream flow Qt = Qr + Q1 + Q2 Qt = 1740 + 28 + 32 = 1800 m³/s BOD of the river downstream of the discharges, BODf = This [BODi* Qi]/Qt = 23 mg/L 21.6*1800 = x= [38880 implies that, (1740*14.8) + (28*248) + (32*x) - 25752 - 6944] / 32 = 193.25 mg/L DO of the river downstream of the discharges, Qi]/Qt DOf = [DOi* = 5.45 mg/ L + 5.45*1800 = (1740*5.6) (28*1.2) + (32*y) y = [9810 - 9744 - 33.6] / 32 = 1.0125 mg/L

Q15 MCQ Water

A treated wastewater discharge enters a river, and complete mixing is assumed to occur immediately downstream of the outfall. The resulting river-water mixture has an ultimate biochemical oxygen demand (BOD) of 15.2 mg/L and a dissolved oxygen concentration of 7.2 mg/L. The saturation dissolved oxygen concentration under the prevailing conditions is 9.4 mg/L. The average flow velocity and mean depth of the stream are 0.35 m/s and 2.8 m, respectively. The deoxygenation and reaeration rate constants are 0.20 day^{-1} and 0.48 day^{-1}, respectively. According to the Streeter–Phelps oxygen sag model, the critical time and the dissolved oxygen concentration corresponding to the critical point are:

A. 2.08 days, 4.0 mg/L

B. 2.54 days, 5.1 mg/L

C. 2.32 days, 5.4 mg/L

D. 2.32 days, 4.0 mg/L

Answer: C

Why: Initial DO deficit, D = DO - DO = 9.4 - 7.2 = 2.2 mg/L Given, L = 15.2 mg/L kd = 0.20 day^{-1} kr = 0.48 day^{-1} Critical time, tc = (1/(kr - kd)) ln [(kr / kd)(1-(D*(kr - kd)/(kd*L)))] = (1/(0.48-0.20))ln[(0.48/0.20)(1-(2.2*(0.48-0.20)/(0.20*15.2)))] = 3.5714ln[2.4(1-0.2026)] = 3.5714*ln(1.9138) = 3.5714*0.6489 = 2.32 days The Streeter–Phelps equation is D = (kd*L/(kr - kd))(e^(-kd*t) - e^(-kr*t)) + D.e^(-kr*t) At the critical point, t = tc = 2.32 days Therefore, Dc = (0.20*15.2/(0.48-0.20))(e^(-0.20*2.32) - e^(-0.48*2.32)) + 2.2e^(-0.48*2.32) = 10.8571*(0.6288-0.3284) + 2.2*0.3284 = 10.8571*0.3004

Q16 MCQ Water

Consider the following statements regarding the dissolved oxygen sag curve: I. Prior to the critical point, the rate of reaeration exceeds the rate of deoxygenation. II. At the critical point, the dissolved oxygen concentration is zero. III. Beyond the critical point, the river begins to recover because the rate of reaeration exceeds the rate of deoxygenation. IV. The self-purification capacity of a river is represented by the DO sag curve. Which of the above statements are correct?

A. I and II only

B. II and III only

C. III and IV only

D. II, III and IV only

Answer: C

Why: Statement I: Incorrect. Prior to the critical point, the rate of deoxygenation exceeds the rate of reaeration, causing the dissolved oxygen deficit to increase. Statement II: Incorrect.

Q17 MCQ Water

Municipal wastewater discharged into a river contains organic nitrogen, which undergoes a series of transformations under aerobic conditions. Which one of the following represents the correct sequence of the predominant nitrogen species with increasing distance downstream from the point of discharge?

A. Organic nitrogen Ammonia nitrogen Nitrite nitrogen Nitrate nitrogen

B. Organic nitrogen Nitrite nitrogen Nitrate nitrogen Ammonia nitrogen

C. Organic nitrogen Ammonia nitrogen Nitrate nitrogen Nitrite nitrogen

D. Organic nitrogen Nitrate nitrogen Nitrite nitrogen Ammonia nitrogen

Answer: A

Why: Organic nitrogen present in wastewater is first converted to ammonia nitrogen by biological decomposition. Under aerobic conditions, Nitrosomonas bacteria oxidize ammonia nitrogen to nitrite nitrogen, which is subsequently oxidized to nitrate nitrogen by Nitrobacter bacteria. Nitrate is the most stable oxidized form of nitrogen.

Q18 MCQ Water

For a river water sample, the quality rating values (q) corresponding to five selected water quality parameters are found to be as follows: Dissolved Oxygen (DO) = 78, Biochemical Oxygen Demand (BOD) = 48, Turbidity = 66, Total Solids = 82, and Nitrates = 34. The respective weighting factors (w) assigned to these parameters are 0.28 for DO, 0.22 for BOD, 0.16 for turbidity, 0.14 for total solids, and 0.20 for nitrates. Using the weighted arithmetic method, determine the Water Quality Index (WQI) of the water sample.

A. 308

B. 67.88

C. 100

D. 61.24

Answer: D

Why: WQI = (wi*qi)/(wi) = ((78 $\blacksquare$ 0.28) + (48 $\blacksquare$ 0.22) + (66 $\blacksquare$ 0.16) + (82 $\blacksquare$ 0.14) + (34 $\blacksquare$ 0.20)) / (0.28 + 0.22 + 0.16 + 0.14 + 0.20) = 61.24

Q19 MCQ Water

Which one of the following applications generally requires tertiary or higher levels of wastewater treatment?

A. Irrigation of fodder and fiber crops

B. Food crop irrigation and unrestricted recreational uses

C. Stream augmentation and habitat conservation

D. Groundwater recharge of non-potable aquifers

Answer: B

Why: Secondary treated wastewater can be used for fodder and fiber crops, groundwater recharge of non-potable aquifers, and stream augmentation. However, applications involving food crop irrigation and unrestricted recreational uses require tertiary or higher treatment involving nutrient removal, filtration, and disinfection.

Q20 MCQ

Water

Consider the following statements regarding wastewater treatment: I. Primary treatment removes dissolved biodegradable organic matter. II. Secondary treatment employs biological processes. III. Conventional wastewater treatment consists of primary and secondary treatment. IV. Disinfection is generally carried out before secondary treatment. Which of the above statements are correct?

A. I and II only

B. I, III and IV only

C. II and III only

D. II, III and IV only

Answer: C

Why: Statement I: Incorrect Primary treatment removes objectionable solids present in wastewater by physical processes such as screening and grit removal. Dissolved biodegradable organic matter, represented by BOD, is removed during secondary treatment. Statement II: Correct Secondary treatment aims at the removal of biodegradable organic matter.

Week 9 — Solid Waste

20 questions · 0 MSQ · 20 other

Q1 MCQ — Solid Waste

In the Integrated Solid Waste Management (ISWM) hierarchy, which of these is the highest priority because it stops waste from being created in the first place?

- **A.** Source reduction and prevention
- **B.** Recycling
- **C.** Waste-to-Energy
- **D.** Landfilling

Answer: A

Why: Reducing resources at the source is the highest priority because it prevents waste from being created, eliminating the need for management.

Q2 MCQ — Solid Waste

What is the reported average Carbon-to-Nitrogen (C/N) ratio for typical Indian urban waste, which is a key factor for bioprocessing?

- **A.** 10.5
- **B.** 33.4
- **C.** 50.2
- **D.** 15

Answer: B

Why: Course data report the average C/N ratio for Indian MSW as about 33.4, which is critical for assessing bioprocessing suitability.

Q3 MCQ — Solid Waste

The Bureau of Indian Standards (BIS) uses a specific symbol for the EcoMark scheme to identify environmentally friendly products. What is it?

- **A.** A green leaf
- **B.** An earthen pot
- **C.** A blue star
- **D.** A recyclable triangle

Answer: B

Why: The earthen pot logo administered by BIS symbolizes the environment-friendly nature of the product under Indian conditions.

Q4 MCQ — Solid Waste

Which of the following heavy metals is NOT part of the "toxic five" identified as having zero beneficial value as nutrients?

A. Mercury (Hg)

B. Iron (Fe)

C. Lead (Pb)

D. Arsenic (As)

Answer: B

Why: As, Cr, Cd, Hg, and Pb are identified as toxic heavy metals with zero beneficial nutrient value. Iron is an essential nutrient.

Q5 MCQ — Solid Waste

Which industry is currently the fastest growing in India, with an annual growth rate between 17% and 21%?

A. Automobiles

B. Electronics

C. Textiles

D. Agriculture

Answer: B

Why: Growing at 17-21% annually, electronics is the fastest-growing industry in India, leading to significant e-waste challenges.

Q6 MCQ — Solid Waste

How many Service Level Benchmarks (SLBs) were set under the 2016 SWM Rules to measure how well a municipality is performing?

A. 5

B. 10

C. 8

D. 12

Answer: B

Why: The SWM Rules 2016 define 10 benchmarks to help Urban Local Bodies measure performance across collection, segregation, and recovery.

Q7 MCQ — Solid Waste

Which thermal process is defined by breaking down waste chemically in the complete absence of air or oxygen?

A. Incineration

B. Gasification

C. Pyrolysis

D. Composting

Answer: C

Why: Unlike incineration (excess air) or gasification (sub-stoichiometric air), pyrolysis occurs in the total absence of air.

Q8 MCQ — Solid Waste

Poor waste management and resulting rodent populations led to a major plague outbreak in 1994 in which Indian city?

A. Mumbai

B. Surat

C. Kolkata

D. Delhi

Answer: B

Why: Open dumping of waste allowed pest species (rodents) to proliferate, leading to the plague in 1994 in Surat.

Q9 MCQ — Solid Waste

Which biological treatment process is specifically anaerobic and produces biogas as a byproduct?

A. Windrow Composting

B. Anaerobic Digestion

C. Vermicomposting

D. Aerobic Digestion

Answer: B

Why: Anaerobic digestion uses bacteria to break down organic matter in a sealed environment without air to produce methane-rich biogas.

Q10 MCQ — Solid Waste

What is the policy called that makes manufacturers responsible for the collection and recycling of their products at the end of their life?

A. Corporate Social Responsibility

B. Extended Producer Responsibility (EPR)

C. Green Tax

D. Product Stewardship

Answer: B

Why: Extended Producer Responsibility mandates that manufacturers take the product back at the end of its life for proper management.

Q11 MCQ — Solid Waste

In a typical Indian municipality, what percentage of the MSW budget is usually spent just on the collection of waste?

A. 10 to 20%

B. 50 to 70%

C. 90 to 100%

D. 30 to 40%

Answer: B

Why: Waste collection is highly labor- and equipment-intensive; course materials note about 50 to 70% of the SWM budget is spent on collection.

Q12 MCQ — Solid Waste

Global frameworks for e-waste (WEEE) officially classify electronic products into how many standard categories?

A. 6 categories

B. 10 categories

C. 3 categories

D. 12 categories

Answer: A

Why: E-waste is standardized into 6 groups: temperature exchange, screens, lamps, large equipment, small equipment, and small IT/telecom.

Q13 MCQ — Solid Waste

Since India lacks natural deposits of Rare Earth Elements (REE), what is the strategy for recovering them from old electronics called?

A. Deep Sea Mining

B. Urban Mining

C. Landfill Mining

D. Chemical Cracking

Answer: B

Why: Since India lacks natural REE deposits, recovering them from discarded electronics (urban mining) helps close the material loop.

Q14 MCQ — Solid Waste

As a country moves from a low-income to a high-income bracket, how does the proportion of paper waste in the MSW stream usually change?

A. It decreases

B. It increases

C. It stays the same

D. It is eliminated

Answer: B

Why: High-income countries consume more packaging and print media, which increases the paper fraction while decreasing the relative organic share.

Q15 MCQ — Solid Waste

An institutional campus at IIT Kharagpur has a residential population of 30,000. During the semester, the 12,000 students generate waste at 0.25 kg/cap.d, and the 18,000 permanent residents generate 0.55 kg/cap.d. Calculate the total daily MSW generated by the entire campus in kg/day.

A. 7,500 kg/day

B. 15,300 kg/day

C. 9,900 kg/day

D. 12,900 kg/day

Answer: D

Why: Students: 12,000 x 0.25 = 3,000 kg/day; residents: 18,000 x 0.55 = 9,900 kg/day; total = 12,900 kg/day.

Q16 MCQ — Solid Waste

An urban local body manages an average daily collection of 1,800 tonnes of unsegregated municipal solid waste. Taking loose bulk density as 450 kg/m^3, calculate the approximate volume capacity (m^3) required per day to transport or temporarily house this daily waste load in its loose form.

A. 2,500 m^3

B. 8,000 m^3

C. 8,100 m^3

D. 4,000 m^3

Answer: D

Why: Daily mass = 1,800 x 1,000 = 1,800,000 kg. Volume = 1,800,000 / 450 = 4,000 m^3.

Q17 MCQ — Solid Waste

A technician analyzes a 50 g dry MSW sample in a muffle furnace at 550 degC. If the sample has 19.63% Volatile Solids on a dry weight basis, what is the final weight of the Fixed Solids (ash) remaining in the crucible in grams?

A. 40.185 g

B. 9.815 g

C. 35.500 g

D. 10.250 g

Answer: A

Why: VS mass = 50 x 0.1963 = 9.815 g; FS mass = 50 - 9.815 = 40.185 g.

Q18 MCQ — Solid Waste

A city's raw MSW has a wet-basis calorific value of 1,530 kcal/kg and 45% moisture. A waste-to-energy plant requires a minimum "as-fired" calorific value of 1,790 kcal/kg. Calculate the maximum moisture content (%) allowed for the waste to meet this threshold, assuming energy is contained entirely in the dry solids.

A. 35.70%

B. 39.80%

C. 32.10%

D. 41.50%

Answer: A

Why: Dry solids calorific value = 1530/0.55 = 2781.8 kcal/kg. Required: 2781.8 x (1-M) = 1790 => M = 0.357 (35.7%).

Q19 MCQ — Solid Waste

A town of 100,000 people generates 0.6 kg/cap.d of waste. The municipality composts 36.7% of generated waste; the rest goes to sanitary landfill compacted to 500 kg/m^3. Calculate the annual landfill volume capacity required in m^3/year.

A. 13,870.3 m^3/year

B. 27,725.4 m^3/year

C. 43,800.7 m^3/year

D. 32,850.9 m^3/year

Answer: B

Why: Annual generation = 100,000 x 0.6 x 365 = 21,900,000 kg. Landfilled (63.3%) = 13,862,700 kg. Volume = 13,862,700/500 = 27,725.4 m^3/year.

Q20 MCQ

Solid Waste

Improper global management of electronic waste accounts for 98 million metric tons of CO2eq refrigerant emissions from 53.6 million metric tons annual generation. India generates 3,230 kt e-waste/year. If a facility captures 36.7% of India's stream, calculate GHG emissions mitigated (million metric tons CO2eq/year).

A. 1.478 million metric tons CO2eq

B. 2.167 million metric tons CO2eq

C. 5.907 million metric tons CO2eq

D. 4.838 million metric tons CO2eq

Answer: B

Why: India mass = 3.23 Mt; captured = 3.23 x 0.367 = 1.185 Mt. Factor = 98/53.6 ? 1.828 tCO2eq/t. Mitigated ? 1.185 x 1.828 = 2.167 Mt CO2eq.

Week 10 — Noise

20 questions · 0 MSQ · 20 other

Q1 MCQ — Noise

Which of the following best defines noise pollution?

A. Any sound louder than 60 dB

B. Unwanted sound that produces undesired physiological or psychological effects in individuals

C. Sound produced only by industrial activities

D. Sound that travels faster than the speed of light

Answer: B

Why: Noise is defined as unwanted sound that produces undesired physiological or psychological effects in individuals. It can arise from traffic, industrial/construction activities, and public events.

Q2 MCQ — Noise

Sound waves are best described as:

A. Transverse electromagnetic waves

B. Alternating waves of compression and rarefaction that cause local changes in air pressure

C. Longitudinal waves that travel only through solids

D. Continuous waves with constant amplitude

Answer: B

Why: Sound waves are alternating waves of compression and rarefaction which result in local changes in air density and therefore pressure. Unlike light, sound waves need a medium to propagate.

Q3 MCQ — Noise

If the speed of sound is c = 340 m/s and the frequency is f = 1000 Hz, what is the wavelength?

A. 0.034 m

B. 0.34 m

C. 3.4 m

D. 34 m

Answer: B

Why: Wavelength = c/f = 340/1000 = 0.34 m.

Q4 MCQ Noise

The reference pressure level for sound pressure level (SPL) measurements is:

A. 1 Pa

B. 200 Pa

C. 20 uPa (20 x 10^-6 Pa)

D. 10^12 Pa

Answer: C

Why: The reference pressure level is 20 microPa (20 x 10^-6 Pa), corresponding to 0 dB and the threshold of human hearing.

Q5 MCQ Noise

The formula for Sound Power Level (Lw) is:

A. Lw = 20 log (W/10^-12)

B. Lw = log (W/10^-12)

C. Lw = 10 log (P^2/P0^2)

D. Lw = 10 log (W/10^-12)

Answer: D

Why: Sound Power Level Lw = 10 log (W/10^-12) where the reference power is 10^-12 W. The factor of 10 (not 20) is used because power is proportional to pressure squared.

Q6 MCQ Noise

If Sound A = 10 dB and Sound B = 20 dB, how many times louder is B compared to A?

A. 2 times

B. 5 times

C. 10 times

D. 20 times

Answer: C

Why: A 10 dB difference represents a factor of 10 in intensity, so B is 10 times louder than A.

Q7 MCQ Noise

What is the sound pressure level (SPL) for two identical sources each at 80 dB?

A. 80 dB

B. 83 dB

C. 160 dB

D. 90 dB

Answer: B

Why: For N identical sources: total SPL = L + 10 log(N) = 80 + 10 log(2) approximately 83 dB. Adding two identical sources increases level by about 3 dB.

Q8 MCQ Noise

For three identical sources each at 60 dB, the combined sound pressure level is approximately:

A. 60 dB

B. 63 dB

C. 180 dB

D. 64.8 dB

Answer: D

Why: Combined SPL = 60 + 10 log(3) approximately 60 + 4.77 = 64.8 dB.

Q9 MCQ Noise

The A-weighting network in a sound level meter:

A. Approximates human response to high intensity sounds

B. Removes low frequencies moderately

C. Approximates human response to low intensity sounds with severe removal of low frequencies

D. Gives equal weight to all frequencies

Answer: C

Why: A-network approximates human response to low intensity sound and removes low frequencies severely. B is for medium intensity; C for high intensity.

Q10 MCQ Noise

A pure tone source emits 90 dB at 100 Hz. What is the expected reading on the A-weighting network if the A-network correction factor at 100 Hz is -19.1 dB?

A. 109.1 dB(A)

B. 90 dB(A)

C. 70.9 dB(A)

D. 84.4 dB(A)

Answer: C

Why: A-network reading = 90 - 19.1 = 70.9 dB(A). At 1000 Hz the correction is 0 for all networks.

Q11 MCQ Noise

At 100 Hz, a source emits 90 dB. The B-network correction is -5.6 dB and C-network correction is -0.3 dB. What are the B and C network readings respectively?

A. 84.4 dB(B) and 89.7 dB(C)

B. 95.6 dB(B) and 90.3 dB(C)

C. 90 dB(B) and 90 dB(C)

D. 70.9 dB(B) and 84.4 dB(C)

Answer: A

Why: B-network = 90 - 5.6 = 84.4 dB(B); C-network = 90 - 0.3 = 89.7 dB(C).

Q12 MCQ Noise

The normal range of human hearing is approximately:

A. 20 Hz to 20,000 Hz

B. 20 Hz to 200,000 Hz

C. 200 Hz to 20,000 Hz

D. 2 Hz to 2000 Hz

Answer: A

Why: Women and children can hear from about 20 Hz to 20,000 Hz. Speech range is roughly 500-2000 Hz; the ear is most sensitive between 2000-5000 Hz.

Q13 MCQ Noise

An octave is defined as:

A. 8 musical notes

B. The frequency interval between a given frequency and one that is triple that frequency

C. The frequency interval between a given frequency and the frequency double that value

D. A range of 1000 Hz

Answer: C

Why: An octave is the frequency interval between a given frequency and twice that frequency (e.g., 22 Hz to 44 Hz).

Q14 MCQ Noise

The geometric mean frequency of the octave band covering 700-1400 Hz is:

A. 1000 Hz

B. 1050 Hz

C. 700 Hz

D. 1400 Hz

Answer: A

Why: Geometric mean = sqrt(700 x 1400) approximately 990 Hz approximately 1000 Hz, the standard octave-band center.

Q15 MCQ Noise

A 'phon' is defined as:

A. The unit of sound power

B. The sound pressure level of a 1000 Hz pure tone in decibels

C. The threshold of hearing in dB

D. The weighting factor at low frequencies

Answer: B

Why: Phon is a psycho-acoustic unit equal to the SPL (in dB) of a 1000 Hz pure tone; equal-loudness contours (Fletcher-Munson) relate loudness across frequencies.

Q16 MCQ — Noise

Temporary Threshold Shift (TTS) in hearing:

A. Is permanent and cannot be recovered

B. Occurs only at sound levels below 40 dB

C. Only affects the C-weighting network response

D. Involves ringing, discomfort or muffling; occurs in first 2 h of exposure with recovery in 16-24 h

Answer: D

Why: TTS involves ringing, discomfort or muffling within the first 2 hours of exposure, with recovery in about 16-24 hours. PTS is the irreversible form.

Q17 MCQ — Noise

Hearing loss due to aging is termed:

A. Presbycusis

B. Tinnitus

C. Temporary Threshold Shift

D. High-frequency notch

Answer: A

Why: Age-related hearing loss is called presbycusis, distinct from noise-induced loss which often shows a notch near 4000 Hz.

Q18 MCQ — Noise

The NIOSH occupational noise exposure limit for an 8-hour workday is:

A. 70 dB(A)

B. 75 dB(A)

C. 85 dB(A)

D. 100 dB(A)

Answer: C

Why: NIOSH considers regular prolonged exposure at or above 85 dB(A) averaged over 8 hours hazardous.

Q19 MCQ — Noise

According to the Inverse Square Law for outdoor sound transmission, if the distance from a point source doubles, the Sound Pressure Level (SPL):

A. Increases by 6 dB

B. Decreases by 6 dB

C. Decreases by 3 dB

D. Remains the same

Answer: B

Why: Intensity falls as 1/r^2; doubling distance reduces SPL by about 6 dB (20 log(1/2) approximately -6 dB).

Q20 MCQ

Noise

Under India's Noise Pollution (Regulation and Control) Rules, 2000, the noise level at the boundary of a public place using a loudspeaker shall not exceed:

A. 5 dB(A) above ambient standards or 75 dB(A), whichever is lower

B. 10 dB(A) above ambient standards or 75 dB(A), whichever is lower

C. 10 dB(A) above ambient standards or 85 dB(A), whichever is lower

D. 15 dB(A) above ambient standards regardless of zone

Answer: B

Why: At the boundary of a public place using loudspeakers/PA systems, noise shall not exceed 10 dB(A) above ambient standards or 75 dB(A), whichever is lower.

Week 11 — Ecology & Soil

20 questions · 4 MSQ · 16 other

Q1 Assertion–Reason — Ecology & Soil

Assertion (A): Compounds having high log Kow values tend to bioaccumulate strongly. Reason (R): Such compounds are more soluble in organic solvents than in water.

- **A.** Both A and R are true and R is the correct explanation of A
- **B.** Both A and R are true but R is not the correct explanation
- **C.** A is true but R is false
- **D.** A is false but R is true

Answer: A

Why: High log Kow values indicate that a chemical is more soluble in fats and organic matter than in water. Such compounds preferentially accumulate in biological tissues, resulting in higher bioaccumulation potential.

Q2 MCQ — Ecology & Soil

An increase in contaminant concentration from minnows predatory fish fish-eating birds is termed:

- **A.** Bioconcentration
- **B.** Bioavailability
- **C.** Biomagnification
- **D.** Biodegradation

Answer: C

Why: Biomagnification is the increase in contaminant concentration at successive trophic levels of a food chain.

Q3 Match — Ecology & Soil

Match List-I with List-II List-I List-II P. Provisioning services 1. Recreation and inspiration Q. Cultural services 2. Oxygen generation R. Supporting services 3. Raw materials and water S. Regulating services 4. Climate control

- **A.** P-3, Q-1, R-2, S-4
- **B.** P-4, Q-3, R-1, S-2
- **C.** P-2, Q-1, R-4, S-3
- **D.** P-3, Q-2, R-1, S-4

Answer: A

Why: Provisioning services provide direct resources from ecosystems such as food, water, and raw materials. Cultural services include recreational and inspirational benefits. Supporting services help sustain life processes like oxygen generation, while regulating services control environmental conditions like climate, therefore.

Q4 MCQ — Ecology & Soil

DDT concentration in lake water is 5 x 10^-5 ppm. If the concentration in herring gull tissue is 6 ppm, then the approximate magnification factor is:

A. 1.2 x 10^4

B. 1.2 x 10^5

C. 1.2 x 10^6

D. 1.2 x 10^3

Answer: B

Why: Magnification Factor = concentration in top predator / concentration in water = 6 / (5 x 10^-5) = 1.2 x 10^5.

Q5 MSQ — select all — Ecology & Soil

Which of the following are examples of regulating ecosystem services?

A. Climate control in wooded areas

B. Flood control

C. Nutrient recycling

D. Disease regulation

Answer: A, B, D

Why: Climate regulation, flood control, and disease regulation help maintain environmental balance and are therefore regulating services. Nutrient recycling is a supporting service, while recreation and tourism belong to cultural services.

Q6 MSQ — select all — Ecology & Soil

Which of the following are true regarding solar radiation and photosynthesis?

A. Infrared radiation forms the largest fraction of solar radiation reaching Earth

B. Chlorophyll absorbs maximum light in the green region

C. UV radiation constitutes a small fraction of incoming radiation

D. Only a fraction of incident solar radiation is usable by plants

Answer: A, C, D

Why: Infrared contributes the largest fraction, UV is a small fraction, and only part of solar radiation can be used in photosynthesis.

Q7 MCQ — Ecology & Soil

The concentration of a pesticide in fish tissue is measured as 3.6 mg/kg. The concentration of the same pesticide in surrounding water is 0.00012 mg/L. The Bioconcentration Factor (BCF) is:

A. 3 x 10^3 L/kg

B. 3 x 10^4 L/kg

C. 300 L/kg

D. 30 L/kg

Answer: B

Why: BCF = 3.6 / 0.00012 = 30,000 = 3 x 10^4 L/kg, indicating strong uptake into fish tissue.

Q8 MCQ — Ecology & Soil

Two species occupying similar habitats but utilizing different food resources illustrate:

A. Identical niches

B. Trophic niche differentiation

C. Competitive exclusion only

D. Ecological succession

Answer: B

Why: Species living in the same habitat but consuming different food resources occupy different trophic niches.

Q9 MCQ — Ecology & Soil

In logistic population growth, the growth rate decreases as population size approaches carrying capacity.

A. TRUE

B. FALSE

Answer: A

Why: Resource limitation increases near carrying capacity (K), reducing growth rate.

Q10 MCQ — Ecology & Soil

A biome characterized by high temperature and very low precipitation is most likely:

A. Tropical rainforest

B. Tundra

C. Sub-tropical desert

D. Temperate forest

Answer: C

Why: Deserts are characterized by high temperature and very low annual rainfall.

Q11 MCQ — Ecology & Soil

A forest ecosystem has a gross primary productivity (GPP) of 2500 kcal m^{2} yr^{2}. If respiration losses account for 35% of GPP, calculate the net primary productivity (NPP).

A. 875 kcal m^{2} yr^{2}

B. 1625 kcal m^{2} yr^{2}

C. 2150 kcal m^{2} yr^{2}

D. 2500 kcal m^{2} yr^{2}

Answer: B

Why: Given: GPP = 2500 kcal m^{2} yr^{2} Respiration = 35% R=0.35^{2}2500=875 NPP=GPP-R=2500-875=1625 kcal m^{2} yr^{2}

Q12 MCQ — Ecology & Soil

In an ecosystem, decomposers mainly help in:

A. Converting solar energy into biomass

B. Recycling nutrients back into the environment

C. Increasing atmospheric oxygen directly

D. Preventing energy loss between trophic levels

Answer: B

Why: Decomposers break down dead organic matter and return nutrients such as nitrogen and phosphorus to the ecosystem.

Q13 MSQ — select all — Ecology & Soil

Which of the following statements regarding ecosystem energy flow are correct?

A. Energy transfer between trophic levels is 100% efficient

B. Solar energy is the primary energy source for most ecosystems

C. Chemical energy stored in biomass moves through trophic levels

D. Nutrients and energy both follow cyclic pathways

Answer: B, C

Why: Solar energy is the primary energy source, and energy stored in biomass moves through trophic levels. Energy is not recycled.

Q14 MSQ — select all — Ecology & Soil

Which processes contribute to loss of incoming solar energy before it reaches Earth's surface?

A. Reflection by clouds

B. Atmospheric absorption

C. Reflection from Earth's surface

D. Biomass respiration

Answer: A, B, C

Why: Cloud reflection, atmospheric absorption, and surface reflection reduce solar radiation before it is utilized at Earth's surface.

Q15 MCQ — Ecology & Soil

Which of the following nutrients is generally considered a major growth-limiting nutrient in agricultural soils?

A. Oxygen

B. Carbon dioxide

C. Nitrogen

D. Hydrogen

Answer: C

Why: Nitrogen is often the primary growth-limiting nutrient because atmospheric nitrogen (N) is not directly bioavailable to plants.

Q16 MCQ — Ecology & Soil

An average soil sample contains: 45% mineral particles, 25% water, 25% air. If 80% of organic matter is humus, determine humus percentage in soil.

A. 20%

B. 8%

C. 4%

D. 25%

Answer: C

Why: Total accounted: 45+25+25=95% Therefore, Organic matter=100-95=5% Humus = 0.8 5=4

Q17 Assertion–Reason — Ecology & Soil

Assertion (A): Overgrazing accelerates soil erosion. Reason (R): Excessive grazing removes vegetation cover and reduces soil cohesion.

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation

C. A is true but R is false

D. A is false but R is true

Answer: A

Why: Overgrazing removes protective vegetation and weakens root binding of soil particles. The exposed soil becomes vulnerable to runoff and wind action, leading to increased soil erosion. Hence both A and R are true, and R correctly explains A.

Q18 MCQ — Ecology & Soil

Which horizon is primarily responsible for maximum accumulation of colloids?

A. O horizon

B. A2 horizon

C. B2 horizon

D. C horizon

Answer: C

Why: B2 horizon is the zone of maximum accumulation of colloidal particles due to downward leaching from upper layers.

Q19 MCQ — Ecology & Soil

Which of the following statements best explains the relationship between soil, vegetation, and climate?

A. Soil determines vegetation, but vegetation does not affect soil

B. Climate affects soil only indirectly through organisms

C. Soil and vegetation influence each other and both depend on climate

D. Vegetation affects climate, but not soil

Answer: C

Why: Soil properties influence vegetation growth, while vegetation modifies soil composition and structure. Both are strongly governed by climatic factors like rainfall and temperature.

Q20 MCQ — Ecology & Soil

Using the Universal Soil Loss Equation: A field has the following parameters: R=250, K=0.3, LS=1.8, C=0.4, P=0.5 Calculate the annual soil loss A. (in tons per unit area per year)

A. (in tons per unit area per year)

B. 108

C. 8.33

D. 15

Answer: A

Why: Universal Soil Loss Equation: A=RKLSCP A=250$\diamond$0.3$\diamond$1.8$\diamond$0.4$\diamond$0.5 A= 27

Week 12 — Policy & Legislation

20 questions · 0 MSQ · 20 other

Q1 MCQ — Policy & Legislation

The Oleum Gas Leak case (1986) is significant primarily because it transformed an existing legal doctrine into a stricter form?

A. The Precautionary Principle into the Polluter Pays Principle

B. The Doctrine of Strict Liability into the Doctrine of Absolute Liability

C. The Public Trust Doctrine into Sustainable Development

D. Criminal Procedure Code liability into Civil liability

Answer: B

Why: Before this case, courts applied 'strict liability' which allowed exceptions. The Supreme Court modified it to 'absolute liability' meaning no exceptions apply when hazardous industries cause harm near human habitation.

Q2 MCQ — Policy & Legislation

In the Ratlam Municipality case (1980), the municipality argued it lacked funds to provide sanitation. What did the Supreme Court decide regarding this defence?

A. It was accepted since financial constraints are a valid legal excuse

B. The case was transferred to the High Court for reassessment

C. The municipality was ordered to act within six months

D. Residents were asked to fund the facilities themselves

Answer: C

Why: The Supreme Court ruled that a municipality cannot escape its statutory duty to provide basic sanitation by pleading financial incapacity.

Q3 MCQ — Policy & Legislation

The Taj Trapezium Zone (TTZ) case involved protecting 255 historic monuments. Which combination of remedies did the Supreme Court order?

A. Shut down the Mathura refinery permanently and plant trees

B. Ban coal use in the zone, switch industries to CNG, and relocate tanneries

C. Impose fines on industries and require them to install filters

D. Create a 10 km no-industry buffer around the Taj Mahal only

Answer: B

Why: The Court ordered a ban on coal, directed industries to shift to Compressed Natural Gas (CNG), asked tanneries to relocate out of the TTZ, and required a green belt around the Mathura refinery.

Q4 MCQ — Policy & Legislation

In the Vellore Citizens Welfare Forum case (1996), tanneries argued they were major foreign exchange earners and employers. How did the Supreme Court balance this against environmental harm?

A. It allowed tanneries to continue operations as economic benefits outweighed harm

B. It transferred enforcement entirely to the Tamil Nadu Pollution Control Board

C. It shut down all tanneries permanently and compensated workers from government funds

D. It applied the Polluter Pays Principle, fined each tannery 10,000, and required them to fund a common effluent treatment plant

Answer: D

Why: The Court acknowledged both sides but upheld the Polluter Pays Principle and Sustainable Development doctrine holding that economic importance could not justify environmental destruction. Each tannery paid a fine and contributed to a Common Effluent Treatment Plant (CETP).

Q5 MCQ — Policy & Legislation

In the Alaknanda Hydropower Project case (NGT, 2016), the company argued the 2013 Uttarakhand floods were an 'Act of God' and it bore no liability. How did the NGT respond?

A. The NGT accepted the Act of God defence and dismissed the petition

B. The NGT ruled the company bore 100% liability since the dam amplified flooding

C. The NGT found that construction muck (approx. 30% of damage) made the company partially liable despite the natural disaster

D. The NGT ordered a fresh environmental clearance before deciding on liability

Answer: C

Why: The NGT applied the No-Fault Liability principle and found that phyllite and construction debris from the project contributed about 30% of the damage, meaning the Act of God defence did not fully shield the company. Each applicant received Rs. 1 lakh compensation.

Q6 MCQ — Policy & Legislation

A rapid EIA of the River Ib Barrage evaluated barrage heights of 4 m, 5 m, and 6 m. Even the tallest option (6 m) was found inadequate. What was the main reason?

A. The 6 m barrage would submerge too many coal mines, violating environmental law

B. The total annual water demand was 94.8 MCM/year, but even the 6 m option could only satisfy about 14.6% of that

C. The Digital Elevation Model showed the site was geologically unstable

D. Evaporation losses exceeded inflow during monsoon months

Answer: B

Why: The estimated water flow into the river was only 65.5 MCM/year, already less than demand. The 6 m barrage met just 14.6% of the 94.8 MCM/year requirement, indicating that a barrage based solution was unlikely to meet the projected demand. Refer Lecture 5.

Q7 MCQ — Policy & Legislation

The Bishnoi movement of 1730 is regarded as one of India's earliest tree protection protests. What was the immediate cause of the conflict in the village of Khejarli?

A. Villagers protested against a new road cutting through their forest

B. King Abhay Singh ordered his men to clear Khejri trees for wood to build his palace

C. A British company obtained a logging permit for the Thar desert region

D. The state government planned to replace natural trees with teak plantations

Answer: B

Why: The Bishnoi community opposed the felling of Khejri trees ordered by King Abhay Singh's officials. Their sacrifice at Khejarli became one of the earliest examples of organized environmental conservation in India.

Q8 MCQ — Policy & Legislation

Which of the following statements regarding the Paris Agreement and India's Nationally Determined Contributions (NDC) is incorrect?

A. India committed to reducing emissions intensity by 33�35% below 2005 levels.

B. India pledged to achieve 40% installed electricity capacity from non fossil fuel sources by 2030.

C. India committed to creating an additional carbon sink of 2.5�3 GtCOe by 2030.

D. The Paris Agreement imposes legally binding emission reduction targets on all countries.

Answer: D

Why: The Paris Agreement is legally binding regarding reporting and NDC submission, but countries determine their own mitigation targets.

Q9 MCQ — Policy & Legislation

Consider the following statements: I. The Vienna Convention primarily focuses on the protection of the ozone layer. II. The Basel Convention deals with transboundary movement and disposal of hazardous waste. III. The Stockholm Convention aims to address climate change and its adverse impacts. IV. The UNFCCC seeks to control persistent organic pollutants (POPs). How many of the above statements are correct?

A. Only one

B. Only two

C. Only three

D. All four

Answer: B

Why: Statements 1 and 2 are correct. Climate change is addressed by the UNFCCC, while the Stockholm Convention deals with POPs.

Q10 MCQ — Policy & Legislation

Under the Biomedical Waste Management Rules (1998), which colour coded container is used for sharp metallic waste such as needles and scalpels, and which is used for broken glassware and medicine vials?

A. Yellow for sharps; red for glassware

B. Red for sharps; white for glassware

C. Blue for sharps; white for glassware

D. White for sharps; blue for glassware

Answer: C

Why: Sharp metallic wastes are collected in blue containers, while broken glassware and medicine vials are placed in white containers to ensure safe segregation and disposal.

Q11 MCQ — Policy & Legislation

Extended Producer Responsibility (EPR) was a key innovation in Indian waste rules. Which of the following rules first clearly defined EPR?

A. Fly Ash Rules 1999

B. Hazardous Waste Rules 1989

C. E Waste Rules 2011 (clarified in 2022)

D. Municipal Solid Waste Rules 2000

Answer: C

Why: EPR places responsibility on producers for managing products after their useful life. It aims to improve recycling, recovery and environmentally sound disposal of electronic waste.

Q12 MCQ — Policy & Legislation

The Doon Valley case (1987) is celebrated as the first Supreme Court judgment to apply a key environmental doctrine in India. Which doctrine was applied, and what was the court's ruling?

A. Polluter Pays Principle; all mining companies were fined and ordered to pay for reforestation

B. Doctrine of Absolute Liability; mine owners were ordered to pay compensation to all residents of Dehradun

C. Public Trust Doctrine; the valley was declared state property and transferred to a government conservation authority

D. Doctrine of Sustainable Development; all mines except three were ordered closed, and economic development was held not to justify environmental degradation

Answer: D

Why: The Supreme Court ruled that economic development should not occur at the cost of environmental destruction. Most limestone mines in the Doon Valley were ordered to shut down.

Q13 MCQ — Policy & Legislation

In M.C. Mehta vs. Kamal Nath (1996), the Supreme Court held that the Himachal Pradesh government had violated an important doctrine by leasing protected forest land on the banks of the Beas River to a motel. Which doctrine was this?

C. Mehta vs. Kamal Nath (1996), the Supreme Court held that the Himachal Pradesh government had violated an important doctrine by leasing protected forest land on the banks of the Beas River to a motel. Which doctrine was this?

A. Doctrine of Absolute Liability

B. Precautionary Principle

D. Doctrine of Sustainable Development

Answer: C

Why: The Court held that natural resources are held by the State in trust for public use and enjoyment. Such resources cannot be transferred for private commercial exploitation.

Q14 MCQ — Policy & Legislation

Which of the following was a major reason for rejecting the proposed vegetable oil industry near Himayat Sagar and Osman Sagar reservoirs?

A. The project was located inside a wildlife sanctuary

B. The project violated coastal regulation norms

C. It posed a potential threat to major drinking water sources

D. The project lacked foreign investment approval

Answer: C

Why: The proposed vegetable oil industry was rejected because it was located near the Himayat Sagar and Osman Sagar reservoirs, which were important sources of drinking water for Hyderabad. The Supreme Court held that the project posed a potential threat to major drinking water reservoirs serving Hyderabad.

Q15 MCQ — Policy & Legislation

The Supreme Court's order requiring cinema halls and television networks to promote environmental awareness originated from:

A. M.

C. Mehta v. Union of India (Environmental Education Case)

B. Vellore Citizens Welfare Forum Case

D. Almitra Patel Case

Answer: A

Why: In M.C. Mehta v. Union of India (Environmental Education Case), the Supreme Court emphasized the importance of environmental awareness among citizens.

Q16 MCQ — Policy & Legislation

Which of the following principles places the burden of proof on the project proponent to demonstrate environmental safety?

A. Polluter Pays Principle

B. Public Trust Doctrine

C. Precautionary Principle

D. Absolute Liability

Answer: C

Why: Under the Precautionary Principle, when there is a potential risk of serious environmental harm, the project proponent must prove that the activity is environmentally safe. The burden of proof lies on the proposer, not on those raising environmental concerns.

Q17 MCQ — Policy & Legislation

The creation of CAMPA (Compensatory Afforestation Fund Management and Planning Authority) emerged from judicial intervention related to:

A. Taj Trapezium conservation

B. Godavarman forest case

C. Oleum gas leakage

D. Ratlam sanitation dispute

Answer: B

Why: CAMPA was established following directions issued in the Godavarman forest case to monitor compensatory afforestation, oversee funds collected for forest diversion, and support the preservation of forests.

Q18 MCQ — Policy & Legislation

Consider the following statements: I. Vembanad﷿Kol Wetland is located in Karnataka. II. The Wildlife (Protection) Act, 1972 falls under the original jurisdiction of the National Green Tribunal. III. A Rapid EIA is always followed by a Comprehensive EIA. How many of the above statements are correct?

A. Only one

B. Only two

C. All three

D. None

Answer: D

Why: Statement 1 is incorrect because Vembanad﷿Kol Wetland is located in Kerala, not Karnataka. Statement 2 is incorrect because the Wildlife (Protection) Act, 1972 is not under the NGT's original jurisdiction. Statement 3 is incorrect because a Rapid EIA is not necessarily followed by a Comprehensive EIA.

Q19 MCQ — Policy & Legislation

A State Government argues that Solid Waste Management Rules cannot be implemented immediately due to inadequate infrastructure and therefore seeks a phased implementation schedule. Which judicial precedent would most directly reject this argument?

A. M.

C. Mehta v. Union of India (Taj Trapezium Case)

B. Vellore Citizens Welfare Forum v. Union of India

D. Almitra H. Patel v. Union of India

Answer: D

Why: In the Almitra H. Patel case, strict enforcement of Solid Waste Management Rules was emphasized. The Tribunal did not support delayed or phased implementation based on local capacity constraints.

Q20 MCQ — Policy & Legislation

Consider the following statements regarding important NGT interventions: I. In the Edwin Wilson case, the NGT took cognizance of the environmental impacts of flex and PVC banners and involved authorities such as the CPCB and State Pollution Control Boards. II. In the Vardhaman Koushik case, the NGT merely advised states to discourage stubble burning in Punjab, Haryana and Uttar Pradesh. III. In both cases, the NGT limited itself to issuing advisory recommendations without directing action by regulatory authorities. How many of the above statements are correct?

A. Only one

B. Only two

C. All three

D. None

Answer: A

Why: Statement 1 is correct. Statement 2 is incorrect because the NGT imposed a ban on stubble burning and issued directions to governments, not merely advisory suggestions. Statement 3 is incorrect because the NGT issued enforceable directions in both cases.

Extra — High-chance / important questions (117)
Predicted exam-style MCQ + MSQ from the course bank. *Practice only — not official NPTEL answer keys.* Green highlights mark the intended correct option set.

High-Likelihood Predicted Questions

Predicted · Week 1

10 high-likelihood questions

p1 MCQ high predicted — formula

In the IPAT identity, which factor is best proxied by 'kg CO2 per kWh of electricity'?

A. Population

B. Affluence

C. Technology

D. Biocapacity

Answer: C

Why: T (technology) captures impact intensity per unit activity; kg CO2/kWh is a classic technology factor.

p2 MCQ high predicted — definition

A substance is detected in soil above natural background but causes no observed harm at that level. Course terminology would call it primarily a:

A. Pollutant

B. Contaminant

C. Criteria pollutant

D. Secondary pollutant

Answer: B

Why: Contaminant = out of place / above background; pollutant implies adverse effects or unreasonable interference.

p3 MCQ high predicted — formula

For the same initial population P0, growth rate r and periods n, which model yields the largest Pn?

A. Arithmetic Pn=P0+nX with X=r*P0

B. Geometric Pn=P0(1+r)^n

C. Exponential Pn=P0 e^(rn)

D. Logistic approaching K

Answer: C

Why: Continuous compounding e^(rn) exceeds discrete (1+r)^n; arithmetic with X=rP0 is linear and smaller for r>0, n>1.

p4 MCQ high predicted concept

Earth Overshoot Day marks the calendar date when:

A. Global CO2 first exceeds 400 ppm

B. Cumulative human demand exceeds what Earth regenerates that year

C. All countries exhaust their national biocapacity

D. EPI of buildings exceeds design limits

Answer: B

Why: Overshoot Day = demand for the year exceeds regenerative biocapacity for that year.

p5 Numerical high predicted numerical

Course global averages compare roughly _____ gha footprint/person to _____ gha biocapacity/person.

A. 1.6 to 2.8

B. 2.8 to 1.6

C. 5.0 to 5.0

D. 0.5 to 3.0

Answer: B

Why: Demand (~2.8 gha) exceeds supply (~1.6 gha) — global overshoot framing.

p6 MCQ high predicted formula

Energy Performance Index (EPI) for a campus is most correctly expressed as:

A. kWh / m^2-year

B. kg CO2 / capita

C. gha / person

D. MJ / GDP

Answer: A

Why: EPI = annual energy use divided by floor area.

p8 Assertion–Reason high predicted definition

Assertion (A): Ecological footprint and biocapacity are both expressed in global hectares. Reason (R): Global hectares adjust land types to world-average productivity so different land uses can be compared.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; gha standardization is why EF and biocapacity share units and can be compared.

p119 Assertion–Reason high predicted formula

Assertion (A): For the same r and n, exponential population growth exceeds geometric growth. Reason (R): Continuous compounding e^(rn) is greater than discrete compounding (1+r)^n for r>0, n>0.

- **A.** Both A and R are true, and R is the correct explanation of A
- **B.** Both A and R are true, but R is not the correct explanation of A
- **C.** A is true, but R is false
- **D.** A is false, but R is true

Answer: A

Why: Both true; e^(rn) > (1+r)^n for positive r,n — classic Week 1 trap.

p121 MSQ — select all high predicted definition

Which of the following are correctly treated as renewable natural resources?

- **A.** Solar radiation
- **B.** Freshwater renewed by the hydrologic cycle
- **C.** Coal seams
- **D.** Wind

Answer: A, B, D

Why: Solar and wind renew continuously; freshwater renews via the hydrologic cycle. Coal is non-renewable.

p122 MSQ — select all high predicted concept

Which combinations indicate an ecological deficit (EF = footprint, BC = biocapacity)?

- **A.** EF = 3, BC = 2
- **B.** EF = 2, BC = 3
- **C.** EF = 5, BC = 4
- **D.** EF = 4, BC = 4

Answer: A, C

Why: Deficit when EF > BC.

Predicted · Week 2

7 high-likelihood questions

p10 MCQ high predicted — definition

The Brundtland (1987) definition of sustainable development emphasizes meeting present needs without:

A. Increasing GDP

B. Compromising future generations' ability to meet their needs

C. Using any non-renewable resources

D. Crossing any planetary boundary immediately

Answer: B

Why: Classic Brundtland one-liner from Our Common Future.

p11 MCQ high predicted — concept

Correct chronological order for sustainability milestones is:

A. SDGs → Rio → Brundtland → MDGs

B. Brundtland 1987 → Rio 1992 → MDGs 2000 → SDGs 2015

C. Rio 1992 → Brundtland 1987 → SDGs 2015 → MDGs 2000

D. MDGs 2000 → Brundtland 1987 → Paris 2015 → Rio 1992

Answer: B

Why: 1987 Our Common Future → 1992 Rio → 2000 MDGs → 2015 SDGs.

p12 MCQ high predicted — concept

Planetary boundaries primarily define:

A. National GDP ceilings

B. A safe operating space for humanity w.r.t. Earth-system processes

C. Maximum renewable energy shares

D. WTO trade quotas

Answer: B

Why: Rockström/Steffen framing: quantified Earth-system limits.

p13 MCQ high predicted — definition

Which SDG number is most directly about climate action?

A. SDG 6

B. SDG 7

C. SDG 12

D. SDG 13

Answer: D

Why: SDG 13 = Climate Action; 6 water, 7 energy, 12 SCP.

p14 MCQ high predicted — concept

Compared with the MDGs, the SDGs are distinctive mainly because they are:

A. Only eight goals for developing countries

B. Seventeen universal goals for all UN Member States

C. Legally binding emission caps for Annex I only

D. Focused solely on poverty metrics

Answer: B

Why: SDGs: 17 goals, universal 2030 Agenda vs MDGs' developing-country focus.

p15 Assertion–Reason high predicted — concept

Assertion (A): The three pillars of sustainability are environment, economy and society. Reason (R): Sustainable development can be achieved by maximizing economic growth alone if technology improves.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: C

Why: A is true (three pillars). R is false — all three pillars must be balanced.

p123 MSQ — select all high predicted — concept

Which of the following are among the classic pillars of sustainable development?

A. Social equity

B. Environmental / ecological integrity

C. Economic viability

D. Military expansion

Answer: A, B, C

Why: Social, environmental, and economic pillars — not militarization.

Predicted · Week 3

11 high-likelihood questions

p17 MCQ high predicted — definition

Tropospheric ozone (O3) is classified as a _____ pollutant because it is _____ .

A. Primary; emitted directly from stacks

B. Secondary; formed by atmospheric reactions of precursors

C. Criteria; only indoor

D. Hazardous; never regulated

Answer: B

Why: Ground-level O3 is classic secondary photochemical pollutant.

p18 MCQ high predicted — law

Which international instrument specifically targets ozone-depleting substances such as CFCs?

A. Kyoto Protocol

B. Paris Agreement

C. Montreal Protocol (1987)

D. UNFCCC alone

Answer: C

Why: Montreal Protocol 1987 = ODS control; do not confuse with climate treaties.

p19 MCQ high predicted — concept

ESP and baghouses are primarily used to control:

A. SO2 only

B. Particulate matter

C. NOx precursors only

D. Indoor radon

Answer: B

Why: Electrostatic precipitators and fabric filters remove particulates (often 95–99%).

p20 MCQ high predicted concept

FGD (flue-gas desulfurization) is applied mainly to reduce emissions of:

A. CO2

B. SO2

C. CH4

D. PM2.5 from cooking

Answer: B

Why: Wet limestone FGD targets sulfur dioxide from combustion.

p21 MCQ high predicted definition

'Ambient air' in the course sense means:

A. Air inside sealed cleanrooms only

B. Outdoor atmospheric air in its natural state that supports life

C. Any indoor kitchen plume

D. Stratospheric ozone layer only

Answer: B

Why: Ambient = outdoor atmospheric air supporting life.

p22 MCQ high predicted definition

Criteria air pollutants are best described as:

A. Rare industrial toxics without standards

B. Common pollutants for which ambient air quality standards are set

C. Only secondary pollutants

D. Only greenhouse gases

Answer: B

Why: Criteria = regulated common ambient pollutants with standards.

p24 Numerical high predicted numerical

Course materials cite indoor/household air pollution mortality on the order of:

A. ~32,000 deaths/year

B. ~3.2 million deaths/year

C. Negligible compared with outdoor

D. Zero after Montreal Protocol

Answer: B

Why: ~3.2 M deaths/y (2020 cite) — major health burden.

p25 Assertion–Reason high predicted concept

Assertion (A): Controlling only primary emissions of NOx and VOCs can help reduce urban ozone. Reason (R): Ozone is emitted directly from vehicle tailpipes as a primary pollutant.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: C

Why: A true (precursor control). R false — O3 is secondary, not directly emitted.

p27 MCQ high predicted definition

Which pair correctly matches primary vs secondary examples?

A. O3 primary; SO2 secondary

B. SO2, NO, CO primary; O3 and secondary PM secondary

C. All PM always secondary

D. CO2 is always a criteria ambient pollutant

Answer: B

Why: SO2/NO/CO/primary PM emitted; O3/secondary PM formed in air.

p124 MSQ — select all high predicted definition

Which of the following are primary precursors of ground-level (tropospheric) ozone?

A. Carbon monoxide only

B. Nitrogen oxides (NOx)

C. Elemental mercury

D. Volatile organic compounds (VOCs)

Answer: B, D

Why: Tropospheric O3 forms from NOx + VOCs in sunlight.

p125 MSQ — select all high predicted concept

Which are ozone-depleting substances (ODSs) targeted in classic Montreal Protocol teaching?

A. CFCs

B. CO2

C. Halons

D. Some chlorinated ODS solvents historically used

Answer: A, C, D

Why: CFCs/halons (and related ODSs) deplete stratospheric ozone; CO2 is mainly a GHG.

Predicted · Week 4

9 high-likelihood questions

p28 MCQ high predicted — definition

Weather differs from climate mainly in:

A. Weather uses only temperature; climate uses only rainfall

B. Weather is short-term state; climate is long-term statistics

C. Climate changes hourly; weather is multi-decadal

D. They are synonyms in IPCC usage

Answer: B

Why: Weather = hours–days; climate = decades+ statistics.

p29 MCQ high predicted — definition

By definition, the Global Warming Potential (GWP) of CO2 is:

A. 0

B. 1

C. 21

D. 310

Answer: B

Why: CO2 is the reference gas: GWP = 1.

p30 MCQ high predicted — concept

Ice–albedo feedback amplifies warming because ice loss tends to:

A. Increase albedo and reflect more sunlight

B. Decrease albedo so more solar energy is absorbed

C. Increase atmospheric N2

D. Stop the greenhouse effect

Answer: B

Why: Darker surfaces absorb more SW → further warming.

p31 MCQ high predicted law

Correct climate-regime timeline is:

A. Paris 1997 → Kyoto 2015 → UNFCCC 2005

B. UNFCCC 1992 → Kyoto 1997 → Paris 2015

C. Kyoto 1992 → UNFCCC 1997 → Paris 2005

D. IPCC 2015 → UNFCCC 1997 → Paris 1992

Answer: B

Why: UNFCCC (Rio 1992), Kyoto 1997 (force 2005), Paris 2015 COP21.

p32 Numerical high predicted numerical

Kyoto Protocol Annex I first-period target cited in course materials is about:

A. 5.2% below 1990 levels

B. 50% below 2005 levels

C. Net-zero by 2030 for all parties

D. No quantified targets

Answer: A

Why: ~5.2% below 1990 for Annex I; Doha later tightened further.

p33 MCQ high predicted concept

Mitigation vs adaptation — which statement is correct?

A. Mitigation only builds sea walls; adaptation cuts emissions

B. Mitigation reduces emissions/enhances sinks; adaptation reduces vulnerability to impacts

C. Adaptation replaces the need for mitigation entirely

D. Neither is needed under Paris

Answer: B

Why: Both required: turn down the stove vs live with residual heat.

p34 MCQ high predicted concept

GHGs are described as largely transparent to _____ but absorbing _____ radiation.

A. IR; UV

B. Shortwave/solar; longwave/IR

C. X-rays; microwaves

D. Visible only; visible only

Answer: B

Why: Transmit SW, trap outgoing IR — greenhouse mechanism.

p35 Assertion–Reason high predicted law

Assertion (A): Under the Paris Agreement, only Annex I countries submit emission pledges. Reason (R): Paris relies on nationally determined contributions (NDCs) from parties.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: D

Why: A false (all parties NDCs). R true — NDC architecture.

p126 MSQ — select all high predicted concept

Which major greenhouse gases also occur naturally in Earth's atmosphere?

A. Carbon dioxide (CO_2)

B. Methane (CH_4)

C. Synthetic CFCs only (no natural source)

D. Water vapor (H_2O)

Answer: A, B, D

Why: CO_2, CH_4, and water vapor occur naturally; CFCs are synthetic.

Predicted · Week 5

10 high-likelihood questions

p38 MCQ high predicted — formula

The Betz limit states that the maximum power coefficient CP of a wind turbine is:

A. 1/2 = 50%

B. 16/27 ≈ 59.3%

C. 8/27 ≈ 29.6%

D. 100%

Answer: B

Why: CP,max = 16/27 ≈ 59.3%; P_max = (8/27)ρAv³.

p39 MCQ high predicted — formula

If wind speed doubles while ρ and A stay constant, available wind power scales by a factor of:

A. 2

B. 4

C. 8

D. 16

Answer: C

Why: P ∝ v³ → 2³ = 8.

p40 MCQ high predicted — definition

Energy trilemma 'SEE' expands to:

A. Solar, Efficiency, Emissions

B. Security, Equity, Environmental sustainability

C. Storage, Export, Electrification

D. Subsidy, Equity, Expansion

Answer: B

Why: World Energy Council framing: Security, Equity, Environment.

p41 MCQ high predicted definition

Renewable energy is defined in the course as energy replenished at a rate:

A. Always zero impact

B. ≥ rate of use by natural processes

C. Only from fossil fuels

D. Only overnight

Answer: B

Why: Replenishment ≥ use — not a claim of zero environmental impact.

p42 MCQ high predicted concept

Compared with solar PV, CSP is more naturally suited to:

A. Diffuse light only with no thermal storage

B. Concentrating heat with optional thermal storage

C. Indoor LED lighting

D. Geothermal flash steam

Answer: B

Why: CSP concentrates heat; thermal storage often integrated.

p43 MCQ high predicted definition

Capacity factor of a wind farm is:

A. Rated power × 8760 always

B. Actual energy produced / energy if run at rated capacity continuously

C. Always equal to Betz limit

D. Air density in kg/m^3

Answer: B

Why: CF = actual / rated-continuous energy over the period.

p44 MCQ high predicted formula

Using $\rho=1.21\ kg/m^3$, $A=\pi(25)^2\ m^2$, $v=10\ m/s$, Betz-limited power is closest to which expression?

A. $(1/2)\rho Av^3$

B. $(8/27)\rho Av^3$

C. ρAv

D. mgz

Answer: B

Why: $P_{wind,max} = (16/27)(1/2)\rho Av^3 = (8/27)\rho Av^3$.

p45 Assertion–Reason high predicted formula

Assertion (A): Real commercial wind turbines often operate below the Betz CP. Reason (R): Betz assumes an idealized actuator disk; real rotors have aerodynamic and mechanical losses, and sites have intermittency reflected in capacity factor.

- **A.** Both A and R are true, and R is the correct explanation of A
- **B.** Both A and R are true, but R is not the correct explanation of A
- **C.** A is true, but R is false
- **D.** A is false, but R is true

Answer: A

Why: Both true; Betz is an upper bound, not a guaranteed operating CP.

p46 MCQ high predicted concept

Intermittency of solar/wind primarily motivates investment in:

- **A.** Only coal baseload forever
- **B.** Energy storage and flexible demand/grid integration
- **C.** Banning all renewables
- **D.** Montreal Protocol revisions

Answer: B

Why: Storage and flexibility bridge supply–demand mismatch.

p128 MSQ — select all high predicted formula

Which factors commonly limit practical solar PV electricity yield at a site?

- **A.** Available land or rooftop area
- **B.** Solar resource / irradiance
- **C.** Module conversion efficiency
- **D.** The Betz limit for wind turbines

Answer: A, B, C

Why: Betz limit applies to wind, not PV.

Predicted · Week 6

10 high-likelihood questions

p47 Numerical high predicted — numerical

Among tidal, wave and OTEC theoretical/technical potentials taught, the largest is:

A. Tidal ~1200 TWh/y

B. Wave ~29500 TWh/y

C. OTEC ~44000 TWh/y

D. All equal

Answer: C

Why: Rank: OTEC ≫ wave ≫ tidal (course numbers).

p48 MCQ high predicted — concept

Enhanced Geothermal Systems (EGS) mainly create value by:

A. Burning biomass underground

B. Stimulating permeability in hot rock with poor natural productivity

C. Capturing wave energy

D. Cooling tropical surface water

Answer: B

Why: Engineer connectivity in hot dry / low-permeability rock.

p49 MCQ high predicted — definition

Fast pyrolysis of biomass is characterized by roughly:

A. 400–600 °C with little/no O2 → bio-oil, gases, char

B. Excess air combustion to ash only

C. 0 °C enzymatic hydrolysis only

D. Nuclear fission of cellulose

Answer: A

Why: Anaerobic thermal decomposition favoring liquids.

p50 MCQ high predicted formula

Deep-water wave power per unit crest width scales primarily with:

A. H only linearly

B. H^2 and period T

C. Air density alone

D. Betz CP

Answer: B

Why: PW ~ (ρ g^2 H^2 T)/32 — height and period dominate.

p51 MCQ high predicted formula

Hydropower efficiency e_hydro is computed as:

A. P_elec / (ṁ g z)

B. ṁ g z / P_elec

C. $(1/2)\rho A v^3$

D. 20 log(P/P0)

Answer: A

Why: Electric output over potential-energy rate of the flow.

p53 MCQ high predicted concept

Open-cycle OTEC differs from closed-cycle mainly because open-cycle:

A. Uses ammonia as working fluid

B. Flash-evaporates seawater as the working vapor

C. Needs no cold deep water

D. Is identical to CSP

Answer: B

Why: Open: seawater vapor; closed: low-BP secondary fluid (e.g. ammonia).

p54 MCQ high predicted formula

Geothermal heat delivery for district heating follows Q =:

A. ṁ cp ΔT

B. 20 log(P/P0)

C. P×A×T

D. R K LS C P

Answer: A

Why: Sensible heat from mass flow and temperature drop.

p55 Assertion–Reason high predicted concept

Assertion (A): Spring tides are typically larger than neap tides. Reason (R): Spring tides occur when Sun–Moon–Earth are aligned so gravitational effects reinforce.

- **A.** Both A and R are true, and R is the correct explanation of A
- **B.** Both A and R are true, but R is not the correct explanation of A
- **C.** A is true, but R is false
- **D.** A is false, but R is true

Answer: A

Why: Both true; alignment → spring; quadrature → neap.

p56 MCQ high predicted formula

Mean wave energy density $E = (1/8)\,\rho\, g\, H^2$ implies energy is most sensitive to errors in:

- **A.** Wave height H
- **B.** Only seawater salinity
- **C.** Betz limit
- **D.** Floor area

Answer: A

Why: $E \propto H^2$ — height measurement errors dominate.

p130 MSQ — select all high predicted concept

Which are potential benefits of biomass pretreatment for biofuel production?

- **A.** Increase surface area for enzymatic hydrolysis
- **B.** Reduce cellulose crystallinity / improve digestibility
- **C.** Help remove or alter lignin barriers
- **D.** Directly finish conversion to retail gasoline with no further steps

Answer: A, B, C

Why: Pretreatment prepares feedstock; it does not itself finish fuel production.

Predicted · Week 7

6 high-likelihood questions

p58 MCQ high predicted — definition

Linear urban metabolism is summarized as:

A. Borrow–Use–Return

B. Take–Make–Waste

C. Reduce–Reuse–Recycle only

D. Photosynthesis–Respiration

Answer: B

Why: One-way throughput depleting natural capital.

p59 MCQ high predicted — concept

Circular economy aims to:

A. Maximize virgin extraction each year

B. Keep materials in use via reuse/repair/remanufacture/recycling

C. Ban all industrial activity

D. Ignore SDG 12

Answer: B

Why: Close loops; design out waste.

p60 MCQ high predicted — definition

Industrial symbiosis means:

A. Competitors never share utilities

B. Waste/by-product of one firm becomes feedstock for another

C. All plants must be identical

D. Only exporting scrap overseas

Answer: B

Why: Ecosystem-like material coupling in industrial clusters.

p61 MCQ high predicted concept

In the urban metabolism metaphor, catabolism corresponds mainly to:

A. Building urban stocks

B. Generating wastes and emissions

C. Photosynthesis

D. Betz limit

Answer: B

Why: Anabolism builds stocks; catabolism yields wastes/emissions.

p62 Assertion–Reason high predicted concept

Assertion (A): Closing material loops can reduce both virgin resource demand and waste emissions. Reason (R): Circular strategies recirculate materials that would otherwise be extracted anew and discarded.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; R explains A.

p132 MSQ — select all high predicted definition

Which strategies align with a circular economy (not linear take-make-dispose)?

A. Product reuse and remanufacturing

B. Design for disassembly and recycling

C. Industrial symbiosis

D. Maximizing single-use disposal without recovery

Answer: A, B, C

Why: Circular strategies keep materials in use.

Predicted · Week 8

8 high-likelihood questions

p64 Numerical high predicted numerical

Roughly what fraction of Earth's water is freshwater (all forms)?

A. 97.5%

B. 2.5%

C. 50%

D. 0%

Answer: B

Why: ~97.5% salt; ~2.5% fresh (tiny usable share).

p65 MCQ high predicted formula

In NSF-WQI weighting, the single highest weight is assigned to:

A. Phosphate

B. Dissolved oxygen

C. Turbidity

D. Hardness

Answer: B

Why: DO weight 0.17 (highest); fecal coliform 0.16 next.

p66 MCQ high predicted definition

Indian BOD convention often emphasized in the course is:

A. BOD5 at 20 °C only worldwide

B. BOD3 at 27 °C

C. COD always equals BOD

D. No oxygen demand parameter

Answer: B

Why: India frequently uses BOD3 @ 27 °C.

p67 MCQ high predicted — definition

Withdrawal differs from consumption because consumption is water that is:

A. Always returned unchanged to the source

B. Not returned (evaporated, incorporated, transpired, etc.)

C. Only rainwater

D. Identical to precipitation

Answer: B

Why: Withdrawals = consumption + returns.

p68 MCQ high predicted — concept

NSF-WQI score band 90–100 is labeled:

A. Very Bad

B. Medium

C. Excellent

D. Unusable

Answer: C

Why: 90–100 Excellent … 0–25 Very Bad.

p69 MCQ high predicted — law

NGT was established on / by Act around:

A. 1974 Water Act only

B. 18 October 2010

C. 1992 Rio

D. 1981 Air Act

Answer: B

Why: NGT Act 2010; est. 18.10.2010; natural justice; ~6-month disposal aim.

p70 Assertion–Reason high predicted — concept

Assertion (A): COD is typically greater than or equal to BOD for the same sample. Reason (R): COD measures chemically oxidizable matter, which can include substances not readily biodegradable in the BOD test.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; COD ≥ BOD usually.

p133 MSQ — select all high predicted definition

Which statements about water withdrawal vs consumption are correct?

A. Withdrawal is water taken from a source

B. Consumption is the portion not returned to the source

C. Withdrawal and consumption are always identical

D. Cooling water returned to a river is largely withdrawn but not fully consumed

Answer: A, B, D

Why: Returned water counts as withdrawal more than consumption.

Predicted · Week 9

8 high-likelihood questions

p72 MCQ high predicted — concept

Correct waste hierarchy order (highest priority first) is:

A. Dispose > Recover > Recycle > Reuse > Prevent

B. Prevent > Reuse > Recycle > Recover > Dispose

C. Recycle > Prevent > Dispose

D. Incinerate > Landfill > Prevent

Answer: B

Why: PRRRD: prevention first, disposal last.

p73 MCQ high predicted — definition

Among the 4Rs, which sits at the top of preference?

A. Recover

B. Recycle

C. Reduce (source reduction)

D. Relocate

Answer: C

Why: Source reduction before reuse/recycle/recover.

p74 MCQ high predicted — concept

Open dumps are unacceptable mainly because they cause:

A. Only visual blight with no emissions

B. Uncontrolled leachate, CH4/VOC emissions, fires and multi-media contamination

C. Perfect LFG capture

D. Zero soil impact

Answer: B

Why: Air, water, soil and explosion risks.

p75 MCQ high predicted — definition

Volatile solids (VS) as a fraction of total solids indicate:

A. Ash content only

B. Organic fraction lost on ignition — relevant to biodegradability

C. Moisture only

D. Metal scrap mass

Answer: B

Why: VS/TS guides biological treatment suitability.

p76 MCQ high predicted — law

E-waste management in India is strongly linked to:

A. Ignoring producer roles

B. E-waste Rules (2011) and Extended Producer Responsibility

C. Montreal Protocol ODS quotas

D. Betz limit

Answer: B

Why: EPR + authorized recyclers under e-waste rules.

p77 Assertion–Reason high predicted — concept

Assertion (A): Sanitary landfills are preferred over open dumps as disposal of last resort. Reason (R): Engineered landfills include liners, cover, leachate and landfill-gas controls that open dumps lack.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; R explains preference within hierarchy.

p78 Numerical high predicted — numerical

A waste sample has 40% moisture. Total solids (TS) as wet-weight fraction is:

A. 40%

B. 60%

C. 100%

D. 0%

Answer: B

Why: TS = 100% − moisture = 60%.

p134 MSQ — select all high predicted concept

Which belong near the top of the ISWM hierarchy (preferred first)?

A. Prevention / reduction at source

B. Reuse

C. Recycling / recovery

D. Uncontrolled open dumping as first choice

Answer: A, B, C

Why: Prevention, reuse, recycling beat dumping.

Predicted · Week 10

14 high-likelihood questions

p79 MCQ high predicted — formula

For a free-field point source, doubling distance changes SPL by about:

A. −3 dB

B. −6 dB

C. −20 dB

D. +6 dB

Answer: B

Why: Inverse square: ΔLp ≈ −20 log(r2/r1) → −6 dB when r2=2 r1.

p80 MCQ high predicted — formula

A busy highway approximated as a line source typically loses about _____ per distance doubling.

A. 6 dB

B. 3 dB

C. 20 dB

D. 0 dB

Answer: B

Why: Line source I ∝ 1/r → ~−3 dB per doubling.

p81 MCQ high predicted — definition

Reference sound pressure P0 defining 0 dB SPL is:

A. 20 Pa

B. 20 µPa

C. 1 atm

D. 0 Pa

Answer: B

Why: P0 = 20 µPa ≈ threshold of hearing.

p82 Numerical high predicted numerical

Two incoherent sources each at 50 dB combine to approximately:

A. 50 dB

B. 53 dB

C. 100 dB

D. 47 dB

Answer: B

Why: Equal levels → +3 dB (energy sum).

p83 MCQ high predicted formula

Lp = 20 log10(Prms/P0). A tenfold increase in rms pressure raises SPL by:

A. 10 dB

B. 20 dB

C. 3 dB

D. 6 dB

Answer: B

Why: 20 log10(10) = 20 dB.

p84 MCQ high predicted formula

Wavelength of sound λ equals:

A. f/c

B. c/f

C. c×f

D. 20 log(c/f)

Answer: B

Why: λ = c/f; period p = 1/f.

p85 MCQ high predicted formula

Approximate speed of sound relation taught is c ≈:

A. 20.05 √T with T in kelvin

B. T/20.05 with T in °C

C. 343 always at any T

D. Independent of temperature

Answer: A

Why: c ≈ 20.05√T (K).

p86 Numerical high predicted numerical

Occupational noise risk caution highlighted is continuous exposure around:

A. 30 dB(A) for 8 h

B. ≥85 dB(A) over 8 h

C. 140 dB(A) conversation

D. 0 dB forever safe regardless of duration

Answer: B

Why: ≥85 dB(A) 8-h occupational risk threshold (course).

p87 MCQ high predicted definition

Leq is best defined as:

A. Instant peak only

B. Equivalent continuous level with same energy as fluctuating noise over the period

C. Night-only C-weighted peak

D. Sound power of the source in watts

Answer: B

Why: Energy-equivalent continuous level.

p88 MCQ high predicted concept

A-weighting is commonly used because it:

A. Ignores human hearing sensitivity

B. Approximates human ear frequency response for environmental/occupational limits

C. Measures only infrasound

D. Converts dB to pascals linearly

Answer: B

Why: A-network mimics ear; dominant for env/occupational limits.

p89 Assertion–Reason high predicted formula

Assertion (A): If two noise levels differ by 10 dB or more, the total is nearly equal to the louder source. Reason (R): On the energy scale, the quieter source contributes negligibly compared with a source 10× higher in intensity (+10 dB).

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; rule of thumb for unequal combination.

p90 Numerical high predicted numerical

A point source measures 70 dB at 10 m. At 40 m (same conditions) the level is nearest:

A. 70 dB

B. 64 dB

C. 58 dB

D. 40 dB

Answer: C

Why: Distance ×4 = two doublings → −12 dB → 58 dB.

p120 Assertion–Reason high predicted formula

Assertion (A): Doubling the distance from a free-field point source reduces SPL by about 6 dB. Reason (R): Intensity falls as 1/r^2, and ΔLp = −20 log10(r2/r1) equals −6 dB when r2=2 r1.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; R is the quantitative explanation of the inverse-square −6 dB rule.

p135 MSQ — select all high predicted definition

Which statements about TTS vs PTS are correct?

A. TTS is temporary hearing sensitivity loss that can recover

B. PTS is lasting hearing damage

C. PTS always recovers fully within minutes

D. Repeated TTS exposures can raise risk of permanent damage

Answer: A, B, D

Why: TTS recovers; PTS does not.

Predicted · Week 11

8 high-likelihood questions

p91 MCQ high predicted — formula

BCF is defined as:

A. C_water / C_biota

B. C_biota / C_water

C. Kow × population

D. Soil loss A

Answer: B

Why: BCF = tissue concentration / water concentration (L/kg).

p92 MCQ high predicted — definition

Biomagnification differs from bioconcentration because biomagnification refers to:

A. Only uptake from water into one organism

B. Increase of concentration across trophic levels

C. Only soil erosion

D. Only acute toxicity tests

Answer: B

Why: Bioconcentration vs water; biomagnification up the food chain.

p93 MCQ high predicted — formula

USLE estimates average annual soil loss A as the product:

A. R × K × L × S × C × P

B. P × A × T

C. $(1/2)\rho Av^3$

D. 20 log(P/P0)

Answer: A

Why: A = R K LS C P (cover & practice are key management levers).

p94 MCQ high predicted formula

Logistic population growth approaches carrying capacity K with form:

A. dN/dt = rN only forever

B. dN/dt = rN(1 − N/K)

C. Pn = P0 + nX only

D. No equilibrium

Answer: B

Why: S-curve; when N≪K behaves like exponential.

p95 MCQ high predicted concept

High log Kow organics tend to:

A. Stay dissolved only in water

B. Bioconcentrate in fatty tissues

C. Have BCF = 0 always

D. Never enter food webs

Answer: B

Why: Hydrophobicity → lipid partitioning → higher BCF.

p96 MCQ high predicted concept

In ecosystems, energy flow is typically _____ while nutrients are _____.

A. Cyclic; unidirectional

B. Unidirectional; cyclic

C. Both always cyclic

D. Neither flows

Answer: B

Why: Energy one-way; mass/nutrients cycle.

p97 Assertion–Reason high predicted concept

Assertion (A): Compounds with high log Kow tend to bioaccumulate strongly. Reason (R): Such compounds are more soluble in organic phases than in water.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; hydrophobicity links Kow to accumulation (course AR pattern).

p136 MSQ — select all high predicted concept

Which are examples of regulating ecosystem services?

A. Climate regulation by forests

B. Flood regulation

C. Nutrient cycling (typically a supporting service)

D. Disease regulation

Answer: A, B, D

Why: Nutrient cycling is classically supporting, not regulating.

Predicted · Week 12

16 high-likelihood questions

p99 MCQ high predicted law

The Oleum Gas Leak case is landmark mainly for establishing:

A. Polluter pays only for municipalities

B. Absolute liability for hazardous industries (no Rylands exceptions)

C. Public trust over Beas banks only

D. NGT procedural timelines

Answer: B

Why: Strict liability hardened to absolute liability — no exceptions for hazardous enterprises.

p100 MCQ high predicted law

In Municipal Council, Ratlam vs Vardhichand, the Court held that lack of municipal funds:

A. Fully excuses sanitation duties

B. Is not a valid defence against statutory public-health/sanitation duties

C. Transfers all liability to the Water Act alone

D. Requires relocating the Taj

Answer: B

Why: Financial incapacity does not nullify obligatory sanitation/public nuisance duties.

p101 MCQ high predicted law

Taj Trapezium Zone (TTZ) remedies included:

A. Only painting the marble

B. Coal ban/CNG shift, tannery relocation, green belt near Mathura refinery

C. Absolute liability for all hotels nationwide

D. Repealing the Air Act

Answer: B

Why: Protect 255 monuments: fuel switch, relocate tanneries, green belt.

p102 MCQ high predicted law

Vellore Citizens Welfare Forum is strongly associated with embedding:

A. Betz limit in statute

B. Polluter pays / precautionary principles in Indian environmental jurisprudence

C. Only Chipko awards

D. EPI building codes

Answer: B

Why: Tanneries vs Palar: development cannot trump environment; polluter pays & precaution.

p103 MCQ high predicted law

M.C. Mehta vs Kamal Nath applied which doctrine to riverbank forest lease for a motel?

A. Absolute liability

B. Public trust doctrine

C. Inverse square law

D. Betz limit

Answer: B

Why: State as trustee of natural resources — public trust doctrine.

p104 MCQ high predicted law

Correct sequence of major Indian statutes is:

A. EPA 1986 → Water 1974 → Air 1981 → NGT 2010

B. Water 1974 → Air 1981 → EPA 1986 → NGT 2010

C. NGT 2010 → Water 1974 → Air 1981 → EPA 1986

D. Air 1974 → Water 1981 → EPA 2010

Answer: B

Why: Water → Air → umbrella EPA → NGT.

p105 MCQ high predicted law

EPA 1986 is best described as:

A. Only a state municipal bye-law

B. Umbrella central law for environmental protection and wide rule-making

C. A treaty identical to Montreal Protocol

D. Repeal of Water and Air Acts

Answer: B

Why: Post-Bhopal-era umbrella; parent of many Rules.

p106 MCQ high predicted — definition

Act vs Rules — correct distinction:

A. Rules state WHAT; Acts state HOW

B. Acts prescribe substantive law (what); Rules detail procedures (how)

C. They are identical documents

D. Rules override the Constitution always

Answer: B

Why: Act = substance; Rules = procedure under the Act.

p107 MCQ high predicted — concept

Chipko movement (Himalaya, from ~1973) is famous for:

A. Urban CNG buses first

B. Non-violent tree-hugging forest protection; Bahuguna marches

C. Oleum absolute liability

D. OTEC plants

Answer: B

Why: Community forest protection; Right Livelihood Award 1987 linked in notes.

p109 MCQ high predicted — law

NGT is guided by _____ and endeavors to dispose of cases within about _____.

A. CPC 1908 strictly; 10 years

B. Natural justice; 6 months

C. Only criminal IPC; 1 day

D. WTO rules; 20 years

Answer: B

Why: Not bound by CPC; natural justice; 6-month endeavor; benches incl. Delhi + four others.

p111 Assertion–Reason high predicted — law

Assertion (A): Hazardous industries near habitation can claim Rylands-style exceptions after the Oleum case.
Reason (R): Oleum transformed the doctrine toward absolute liability without those exceptions.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: D

Why: A false; R true — absolute liability eliminates traditional exceptions.

p112 Assertion–Reason high predicted law

Assertion (A): Under the public trust doctrine, the State holds certain natural resources in trust for the public. Reason (R): M.C. Mehta vs Kamal Nath applied this doctrine to a forested Beas riverbank lease.

A. Both A and R are true, and R is the correct explanation of A

B. Both A and R are true, but R is not the correct explanation of A

C. A is true, but R is false

D. A is false, but R is true

Answer: A

Why: Both true; Kamal Nath is the course exemplar.

p113 MCQ high predicted concept

EIA in the Week 12 framing is primarily a tool to:

A. Predict/evaluate impacts before project decisions and attach clearance conditions

B. Replace all Acts with voluntary codes

C. Measure only workplace Leq

D. Compute Betz CP

Answer: A

Why: Predictive, participatory assessment linked to clearances.

p117 MCQ high predicted law

Water Act 1974's institutional contribution includes creation/empowerment of:

A. Only NGT

B. CPCB/SPCBs for water pollution prevention/control

C. Only municipal corporations

D. IPCC Working Groups

Answer: B

Why: Boards architecture for water pollution control.

p118 MCQ high predicted law

Air Act year (original) is:

A. 1974

B. 1981 (amended 1987)

C. 1986

D. 2010

Answer: B

Why: Air 1981, am. 1987; EPA 1986; Water 1974; NGT 2010.

p138 MSQ — select all high predicted law

Which legal ideas are strongly associated with landmark Indian environmental cases in this course?

- **A.** Absolute liability (Oleum Gas Leak)
- **B.** Polluter Pays (e.g., Vellore)
- **C.** Public Trust Doctrine (e.g., Kamal Nath)
- **D.** Unlimited right to pollute if any tax is paid

Answer: A, B, C

Why: Oleum, Vellore, and Kamal Nath map to absolute liability, polluter pays, and public trust.